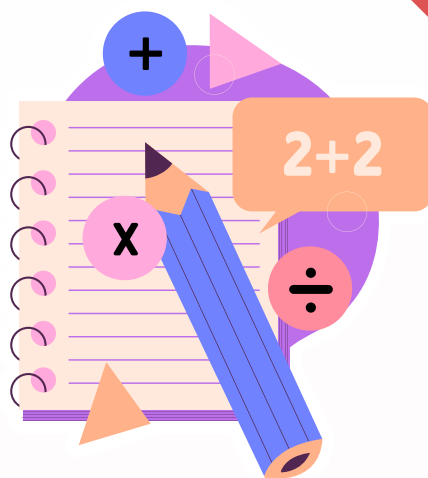


JEE (MAIN+ADVANCED)

Mathematics

ENTHUSIAST COURSE



STUDY MATERIAL

Indefinite & Definite Integration

English Medium

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$$\begin{aligned}
 \text{(vii)} \quad \int \tan(ax+b)dx &= \frac{1}{a} \ell n |\sec(ax+b)| + C & \text{(viii)} \quad \int \cot(ax+b)dx &= \frac{1}{a} \ell n |\sin(ax+b)| + C \\
 \text{(ix)} \quad \int \sec^2(ax+b)dx &= \frac{1}{a} \tan(ax+b) + C \\
 \text{(x)} \quad \int \operatorname{cosec}^2(ax+b)dx &= -\frac{1}{a} \cot(ax+b) + C \\
 \text{(xi)} \quad \int \operatorname{cosec}(ax+b) \cdot \cot(ax+b)dx &= -\frac{1}{a} \operatorname{cosec}(ax+b) + C \\
 \text{(xii)} \quad \int \sec(ax+b) \cdot \tan(ax+b)dx &= \frac{1}{a} \sec(ax+b) + C \\
 \text{(xiii)} \quad \int \sec x dx &= \ell n |\sec x + \tan x| + C = \ell n \left| \tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right| + C \\
 \text{(xiv)} \quad \int \operatorname{cosec} x dx &= \ell n |\operatorname{cosec} x - \cot x| + C = \ell n \left| \tan \frac{x}{2} \right| + C = -\ell n |\operatorname{cosec} x + \cot x| + C \\
 \text{(xv)} \quad \int \frac{dx}{\sqrt{a^2 - x^2}} &= \sin^{-1} \frac{x}{a} + C & \text{(xvi)} \quad \int \frac{dx}{a^2 + x^2} &= \frac{1}{a} \tan^{-1} \frac{x}{a} + C \\
 \text{(xvii)} \quad \int \frac{dx}{x\sqrt{x^2 - a^2}} &= \frac{1}{a} \sec^{-1} \frac{x}{a} + C & \text{(xviii)} \quad \int \frac{dx}{\sqrt{x^2 + a^2}} &= \ell n \left[x + \sqrt{x^2 + a^2} \right] + C \\
 \text{(xix)} \quad \int \frac{dx}{\sqrt{x^2 - a^2}} &= \ell n \left[x + \sqrt{x^2 - a^2} \right] + C & \text{(xx)} \quad \int \frac{dx}{a^2 - x^2} &= \frac{1}{2a} \ell n \left| \frac{a+x}{a-x} \right| + C \\
 \text{(xxi)} \quad \int \frac{dx}{x^2 - a^2} &= \frac{1}{2a} \ell n \left| \frac{x-a}{x+a} \right| + C & \text{(xxii)} \quad \int \sqrt{a^2 - x^2} dx &= \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C \\
 \text{(xxiii)} \quad \int \sqrt{x^2 + a^2} dx &= \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \ell n \left(x + \sqrt{x^2 + a^2} \right) + C \\
 \text{(xxiv)} \quad \int \sqrt{x^2 - a^2} dx &= \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ell n \left(x + \sqrt{x^2 - a^2} \right) + C \\
 \text{(xxv)} \quad \int e^{ax} \cdot \sin bx dx &= \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C = \frac{e^{ax}}{\sqrt{a^2 + b^2}} \sin \left(bx - \tan^{-1} \frac{b}{a} \right) + C \\
 \text{(xxvi)} \quad \int e^{ax} \cdot \cos bx dx &= \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C = \frac{e^{ax}}{\sqrt{a^2 + b^2}} \cos \left(bx - \tan^{-1} \frac{b}{a} \right) + C
 \end{aligned}$$

Elementary Integrals (Loving Integrals)

(Direct formulae Based or Converting into Known formulae)

Illustration 1 : $\int (x+5)^2 dx$

Solution : **M-1 :** $\int (x^2 + 25 + 10x) dx = \frac{x^3}{3} + \frac{10x^2}{2} + 25x + C$

M-2 : $= \frac{(x+5)^3}{3} + C = \frac{x^3 + 15x^2 + 75x + 125}{3} + C$

Illustration 2 : $\int \frac{x^4 + 1}{x^3} dx$

Solution : $\int x^1 + \frac{1}{x^3} dx = \frac{x^2}{2} + \frac{x^{-2}}{(-2)} + C = \frac{x^2}{2} - \frac{1}{2x^2} + C$

Do yourself - 1 :

1. $\int 2^x \cdot e^x dx$

2. $\int (e^{a \ell nx} + e^{x \ell na}) dx \quad (a > 0)$

3. $\int \frac{e^{5 \ell nx} - e^{4 \ell nx}}{e^{3 \ell nx} - e^{2 \ell nx}} dx$

4. $\int \frac{x^6 - 1}{x^2 + 1} dx$

5. $\int \frac{1 + \cos^2 x}{1 + \cos 2x} dx$

6. $\int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx$

7. $\int \frac{\sec 2x - 1}{\sec 2x + 1} dx$

8. $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx \quad (\cos x + \sin x > 0)$

9. $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx$

10. $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx$

11. $\int \frac{\cos^4 x - \sin^4 x}{\sqrt{1 + \cos 4x}} dx \quad (\cos 2x > 0)$

12. $\int \frac{1 + \tan^2 x}{1 + \cot^2 x} dx$

13. $\int \frac{1 + 2x^2}{x^2(1 + x^2)} dx$

14. $\int \frac{(1 + x)^2}{x(1 + x^2)} dx$

15. $\int \frac{2x - 1}{x - 2} dx$

16. $\int \frac{2 + 3x^2}{x^2(1 + x^2)} dx$

17. $\int \frac{x^5 + 3x^4 - x^3 + 8x^2 - x + 8}{x^2 + 1} dx$

18. $\int \frac{x^4 + x^2 + 1}{2(1 + x^2)} dx$

19. $\int \frac{(x^2 + \sin^2 x) \sec^2 x}{1 + x^2} dx$

20. $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x} dx$

21. $\int \frac{\sin 2x + \sin 5x - \sin 3x}{\cos x + 1 - 2 \sin^2 2x} dx$

22. $\int \frac{dx}{1 + \sin x}$

23. $\int \frac{\sin 2x - \sin 2k}{\sin x - \sin k + \cos x - \cos k} dx$

24. $\int \frac{dx}{1 - \cos x}$

Extension of standard formulae :

If x is replaced by $(Ax + B)$ then also the standard result remain true provided the result is divided by co-efficient of x

$$\int (Ax + B)^n dx = \frac{(Ax + B)^{n+1}}{(n+1)A} + C, n \neq -1$$

Similarly if $\int f(x) dx = g(x) + C$ then $\int f(Ax + B) dx = \frac{g(Ax + B)}{A} + C$

Illustration 3 : Evaluate (i) $\int \frac{dx}{8x+3}$ (ii) $\int e^{8x+9} dx$

Solution : (i) $\frac{1}{8} \ln|8x+9| + C$ (ii) $\frac{e^{8x+9}}{8} + C$

Illustration 4 : Evaluate (i) $\int \cos(6+2x) dx$ (ii) $\int \sec^2(6-5x) dx$

Solution : (i) $\frac{\sin(5+4x)}{4} + C$ (ii) $\frac{\tan(3-4x)}{-4} + C$

Illustration 5 : $\int \cos^2 x dx$

Solution : $\int \frac{1+\cos 2x}{2} dx = \frac{x}{2} + \frac{\sin 2x}{2 \cdot 2} + C$

Illustration 6 : $\int \cos^3 x dx$

Solution : $\int \frac{3\cos x + \cos 3x}{4} dx$ [$\cos 3x = 4\cos^3 x - 3\cos x$]

Illustration 7 : $\int \sin^4 x dx$

Solution :

$$\begin{aligned} & \int (\sin^2 x)^2 dx \\ &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 dx \\ &= \int \frac{1 + \cos^2 2x - 2\cos 2x}{4} dx \\ &= \int \frac{1}{4} dx + \int \frac{\cos^2 2x}{4} dx - \frac{2}{4} \int \cos 2x dx \\ &= \int \frac{1}{4} dx + \int \frac{1 + \cos 4x}{2 \cdot 4} dx - \int \frac{2\cos 2x}{4} dx \\ &= \frac{x}{4} + \frac{1}{8} x + \frac{1}{8} \cdot \frac{\sin 4x}{4} - \frac{1}{2} \cdot \frac{\sin 2x}{2} + C \\ &= \frac{3}{8} x + \frac{1}{32} \sin 4x - \frac{1}{4} \sin 2x + C \end{aligned}$$

Illustration 8: $\int \tan 3x \tan 2x \tan x \, dx$

Solution : $3x = x + 2x$

$$\tan 3x = \tan(x + 2x)$$

$$\tan 3x = \frac{\tan x + \tan 2x}{1 - \tan x \tan 2x}$$

$$\tan^3 x - \tan x \cdot \tan^2 x \cdot \tan^3 x = \tan x + \tan^2 x$$

$$\tan x \tan 2x \tan 3x = \tan 3x - \tan x - \tan 2x$$

$$\int (\tan 3x - \tan 2x - \tan x) dx$$

$$\int \tan 3x \, dx - \int \tan 2x \, dx - \int \tan x \, dx$$

$$\frac{\ln|\sec 3x|}{3} - \frac{\ln|\sec 2x|}{2} - \ln|\sec x| + C$$

Do yourself - 2 :

1. $\int e^{-3x+1} dx$

2. $\int (x + 1)^{15} dx$

3. $\int \frac{e^{2x} - 1}{e^x} dx$

4. $\int \frac{dx}{(2x-3)^5}$

5. $\int \frac{dz}{(a+bz)^c} (c \neq 1)$

6. $\int \sqrt[5]{(8-3x)^6} dx$

7. $\int \sqrt{8-2x} dx$

8. $\int \frac{m}{\sqrt[3]{(a+bx)^2}} dx$

9. $\int \frac{2x+3}{3x+2} dx$

10. $\int (e^x + 1)^3 dx$

11. $\int \cos^2 x dx$

12. $\int \sin^2 x \, dx$

13. $\int \frac{1 - \cos x}{1 + \cos x} dx$

14. $\int \cos x \sin 3x dx$

15. $\int (\cos \alpha - \cos 2x) dx$

16. $\int \left[\cos \left(2x - \frac{\pi}{4} \right) \right]^{-2} dx$

17. $\int \cos x \cos 2x \cos 3x \, dx$

18. $\int \frac{d\phi}{\sin^2 \phi \cos^2 \phi}$

19. $\int \cos^4 x \, dx$

20. $\int \frac{1 - \tan^2 x}{1 + \tan^2 x} dx$

21. $\int \frac{\cos x - \sin x}{\cos x + \sin x} (2 + 2 \sin 2x) dx$

22. $\int (3 \sin x \cos^2 x - \sin^3 x) dx$ **23.** $\int \cos x^\circ dx$

24. $\int \sin x \cos x \cos 2x \cos 4x \, dx$

25. $\int \left[\sin^2 \left(\frac{9\pi}{8} + \frac{x}{4} \right) - \sin^2 \left(\frac{7\pi}{8} + \frac{x}{4} \right) \right] dx$

26. $\int \frac{\cos 8x - \cos 7x}{1 + 2\cos 5x} dx$

27. $\int 4 \cos \frac{x}{2} \cdot \cos x \cdot \sin \frac{21}{2} x dx$

28. $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx$

29. $\int \left[\frac{\cot^2 2x - 1}{2 \cot 2x} - \cos 8x \cot 4x \right] dx$

30. $\int \frac{2x^3 + 3x^2 + 4x + 5}{2x + 1} dx$

3. METHODS OF INTEGRATION :

(a) Substitution or change of independent variable :

If $\phi(x)$ is a continuous differentiable function, then to evaluate integrals of the form $\int f(\phi(x))\phi'(x)dx$, we substitute $\phi(x) = t$ and $\phi'(x)dx = dt$.

Hence $I = \int f(\phi(x))\phi'(x)dx$ reduces to $\int f(t)dt$.

(i) Fundamental deductions of method of substitution :

$$\int [f(x)]^n f'(x)dx \quad \text{OR} \quad \int \frac{f'(x)}{[f(x)]^n} dx \quad \text{put } f(x) = t \text{ \& proceed.}$$

Substitution is said to be appropriate if the integrand in (i) is a loving one.

$$\text{If } \int f(x)dx = \phi(x)$$

$$\int f(g(x))g'(x)dx = ? \quad \dots(i)$$

$$\text{Let } g(x) = t \Rightarrow g'(x)dx = dt$$

$$\begin{aligned} \int f(t)dt &= \phi(t) + c \\ &= \phi(t) + c \end{aligned}$$

When substitution is directly observed :

Illustration 9 : $\int \frac{\ln x^2}{x} dx$

Solution : Let $\ln x^2 = t$

$$\frac{1}{x^2} \cdot 2x dx = dt$$

$$\frac{2dx}{x} = dt$$

$$\Rightarrow \int \frac{t}{2} dt = \frac{t^2}{4} + C = \frac{(\ln x^2)^2}{4} + C$$

Illustration 10 : $\int \frac{\sin^{-1}x}{\sqrt{1-x^2}} dx$

Solution : Let $\sin^{-1}x = t$

$$\int \frac{1}{\sqrt{1-x^2}} dx = dt$$

$$\Rightarrow \int t dt = \frac{t^2}{2} + C \Rightarrow (\sin^{-1}x)^2 + C$$

Illustration 11 : $\int \frac{\sin(\tan^{-1}x)}{1+x^2} dx$

Solution : Put, $\tan^{-1}x = t$

$$\frac{1}{1+x^2} dx = dt$$

$$\Rightarrow \int \sin t dt = -\cos t + C = -\cos(\tan^{-1}x) + C$$

Illustration 12 : $\int (4x+6)\sqrt{2x^2+6x+5} \, dx$

Solution : Let $2x^2 + 6x + 5 = t$
 $(4x+6)dx = dt$

$$\Rightarrow \int \sqrt{t} dt = \frac{t^{3/2}}{3/2} + C$$

$$= \frac{2}{3}(2x^2 + 6x + 5)^{3/2} + C$$

Do yourself - 3 :

1. Integrate with respect to x :

- | | | |
|--|--|---|
| (i) $x \sin x^2$ | (ii) $\frac{e^x + 1}{e^x + x}$ | 4. $\int \frac{x dx}{x^2 + 1}$ |
| 2. $\int \frac{d(1+x^2)}{1+x^2}$ | 3. $\int \frac{(2x-3)dx}{x^2-3x+8}$ | 7. $\int \left(\frac{x}{x+1} - \ell n(x+1) \right) dx$ |
| 5. $\int \frac{\sqrt{\ell nx}}{x} dx$ | 6. $\int \frac{(6x-5)dx}{2\sqrt{3x^2-5x+6}}$ | 10. $\int \frac{dx}{x \ell nx}$ |
| 8. $\int \frac{x^2 dx}{x^3 + 1}$ | 9. $\int \frac{e^x dx}{e^x + 1}$ | 13. $\int e^{-x^3} x^2 dx$ |
| 11. $\int \frac{(\ell nx)^m}{x} dx$ | 12. $\int e^{x^2} x \, dx$ | 16. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$ |
| 14. $\int \frac{d\left(\frac{x}{3}\right)}{\sqrt{1-\left(\frac{x}{3}\right)^2}}$ | 15. $\int (x+1)\sqrt{x^2+2x} \, dx$ | 19. $\int e^{e^x+x} dx$ |
| 17. $\int \frac{2x^5-3x^2}{1+3x^3-x^6} dx$ | 18. $\int \frac{\sqrt{x} dx}{1+x^2}$ | 22. $\int \tan^3 x d(\tan x)$ |
| 20. $\int e^{2x^2+\ell nx} dx$ | 21. $\int \sin x d(\sin x)$ | 25. $\int \sqrt{1-e^x} e^x dx$ |
| 23. $\int \frac{d(1+x^2)}{\sqrt{1+x^2}}$ | 24. $\int \frac{(1+x)^2}{x^2+1} dx$ | 28. $\int \tan 3x \, dx$ |
| 26. $\int \frac{d(1+\ell nx)}{\cos^2(1+\ell nx)}$ | 27. $\int \frac{d(\arcsin x)}{\arcsin x}$ | 31. $\int 2x\sqrt{x^2+1} \, dx$ |
| 29. $\int \cot(2x+1) \, dx$ | 30. $\int e^{\sin x} d(\sin x)$ | 34. $\int \frac{e^{2x} dx}{e^{2x} + a^2}$ |
| 32. $\int x\sqrt{1-x^2} \, dx$ | 33. $\int x^2 \sqrt[5]{x^3+2} \, dx$ | 37. $\int \frac{x^3 dx}{\sqrt[3]{x^4+1}}$ |
| 35. $\int \frac{(\arctan x)^n}{1+x^2} dx$ | 36. $\int \frac{x^4 dx}{\sqrt{4+x^5}}$ | |

Converting the given integral form to make it suitable for substitution or to get directly loving form :

Illustration 13 : $\int \frac{\sqrt{\tan x}}{\sin 2x} dx$

Solution : $\int \frac{\sqrt{\tan x}}{2 \tan x} (1 + \tan^2 x) dx$

$$\Rightarrow \int \frac{\sqrt{\tan x}}{2 \tan x} \sec^2 x dx$$

$$\tan x = t$$

$$\sec^2 x = \frac{dt}{dx}$$

$$\frac{1}{2} \int (t)^{-1/2} dt = \frac{t^{1/2} \times 2}{2 \times 1} + C$$

$$= \sqrt{t} = \sqrt{\tan x} + C$$

Illustration 14 : $\int \frac{2x^3 dx}{1+x^2}$

Solution : Put $1+x^2 = t$
 $2x dx = dt$

$$\Rightarrow \int \frac{2x \cdot x^2 dx}{1+x^2} = \int \frac{(t-1)}{t} dt$$

$$= \int \left(1 - \frac{1}{t}\right) dt$$

$$= t - \ln t + C$$

$$= (1+x^2) - \ln(1+x^2) + C$$

Illustration 15 : $\int \frac{9x^2 + 2}{3x^3 + 2x + 10} dx$

Solution : Put $3x^3 + 2x + 10 = t$
 $(9x^2 + 2) dx = dt$

$$= \int \frac{dt}{t} = \ln t + C$$

$$= \ln(3x^3 + 2x + 10) + C$$

Illustration 16 : Evaluate $\int \frac{(x^2 - 1) dx}{(x^4 + 3x^2 + 1) \tan^{-1}\left(x + \frac{1}{x}\right)}$

Solution : The given integral can be written as

$$I = \int \frac{\left(1 - \frac{1}{x^2}\right) dx}{\left[\left(x + \frac{1}{x}\right)^2 + 1\right] \tan^{-1}\left(x + \frac{1}{x}\right)}$$

Let $\left(x + \frac{1}{x}\right) = t$. Differentiating we get $\left(1 - \frac{1}{x^2}\right) dx = dt$

$$\text{Hence } I = \int \frac{dt}{(t^2 + 1) \tan^{-1} t}$$

Now make one more substitution $\tan^{-1} t = u$. Then $\frac{dt}{t^2 + 1} = du$ and $I = \int \frac{du}{u} = \ln |u| + C$

Returning to t , and then to x , we have

$$I = \ln |\tan^{-1} t| + C = \ln \left| \tan^{-1} \left(x + \frac{1}{x} \right) \right| + C$$

Ans.

Do yourself - 4 :

1. $\int e^x (\sin e^x) dx$

2. $\int \frac{x}{\sqrt{x+2}} dx$

3. $\int \frac{e^x - 1}{e^x + 1} dx$

4. $\int x^x \ln(ex) dx$

5. $\int \frac{x^3 dx}{\sqrt{1+x^2}}$

6. $\int \frac{x}{2x+1} dx$

7. $\int \frac{x^4}{1-x} dx$

8. $\int \frac{x^3 dx}{x+1}$

9. $\int \frac{dx}{1+\sqrt{x+1}}$

10. $\int \frac{dx}{1+\sqrt[3]{x+1}}$

11. $\int \frac{(x + \sqrt{1+x^2})^{15}}{\sqrt{1+x^2}} dx$

12. $\int \frac{(\sqrt{x} + 1)(x^2 - \sqrt{x})}{x\sqrt{x} + x + \sqrt{x}} dx$

13. $\int \frac{(\sqrt{x} + 1) dx}{\sqrt{x}(\sqrt[3]{x} + 1)}$

14. $\int \frac{\cot x}{\ln(\sin x)} dx$

15. $\int \frac{dx}{x^4 \sqrt{x^2 + 4}}$

16. $\int \frac{dx}{x^4 \sqrt{x^2 - 3}}$

17. $\int \frac{dx}{(x + \sqrt{x^2 - 1})^2}$

Substitutions involving trigonometric functions :

- $\int \cot x dx = \int \frac{\cos x}{\sin x} dx = \ln |\sin x| + C$
- $\int \tan x dx = - \int \frac{\sin x}{\cos x} dx = - \ln |\cos x| + C \text{ or } \ln |\sec x| + C$
- $\int \sec x dx = \int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} dx = \ln |\sec x + \tan x| + C \text{ or } \ln \tan \left(\frac{x}{2} + \frac{\pi}{4} \right) + C$
- $\int \operatorname{cosec} x dx = \int \frac{\operatorname{cosec} x (\cot x - \operatorname{cosec} x)}{\cot x - \operatorname{cosec} x} dx = \ln |(\cot x - \operatorname{cosec} x)| + C \text{ or } \ln \left| \tan \frac{x}{2} \right| + C$

Illustration 17: Evaluate $\int \frac{\cos^3 x}{\sin^2 x + \sin x} dx$

Solution : $I = \int \frac{(1 - \sin^2 x) \cos x}{\sin x (1 + \sin x)} dx = \int \frac{1 - \sin x}{\sin x} \cos x dx$
 Put $\sin x = t \Rightarrow \cos x dx = dt$
 $\Rightarrow I = \int \frac{1-t}{t} dt = \ln |t| - t + C = \ln |\sin x| - \sin x + C$

Ans.

Illustration 18 : $\int \frac{e^{2x}(1+2x)}{\sin(xe^{2x})} dx$

Solution : Put, $xe^{2x} = t$
 $e^{2x} + 2xe^{2x} dx = dt$
 $e^{2x}(1+2x) dx = dt$
 $\Rightarrow \int \frac{dt}{\sin t} = \int \operatorname{cosec} t dt$
 $= \ln \left| \tan \frac{t}{2} \right| + C$
 $= \ln \left| \tan \frac{xe^{2x}}{2} \right| + C$

Illustration 19 : $\int \frac{\cos 2x}{\cos x} dx$

Solution : $\int \frac{2\cos^2 x - 1}{\cos x} dx$
 $\Rightarrow \int (2\cos x - \sec x) dx$
 $\Rightarrow 2\sin x - \ln |\sec x + \tan x| + C$

Illustration 20 : $\int \operatorname{cosec} x \ln |\cot x - \operatorname{cosec} x| dx$

Solution : Put $\operatorname{cosec} x = t$

$$\ln|\cot x - \operatorname{cosec} x| dx = dt$$

$$\Rightarrow \int t dt = \frac{t^2}{2} + C$$

$$= \frac{\operatorname{cosec}^2 x}{2} + C$$

Illustration 21 : $\int \frac{dx}{\sin(x+5)\sin(x+8)}$

Solution : Multiplying & dividing by $\sin(8-5)$

$$\Rightarrow \frac{1}{\sin(8-5)} \int \frac{\sin(8-5)}{\sin(x+5)\sin(x+8)} dx$$

$$\Rightarrow \frac{1}{\sin 3} \int \frac{\sin((x+8)-(x+5)) dx}{\sin(x+5)\sin(x+8)}$$

$$\Rightarrow \frac{1}{\sin 3} \int \frac{\sin(x+8)\cos(x+5) - \cos(x+8)\sin(x+5)}{\sin(x+5)\sin(x+8)} dx$$

$$\Rightarrow \frac{1}{\sin 3} \int (\cot(x+5) - \cot(x+8)) dx$$

Do yourself - 5 :

1. (a) $\int \sin^3 x \cos x dx$ (b) $\int \frac{\sin x dx}{\cos^2 x}$

(b) $\int \frac{\sin x \, dx}{\cos^2 x}$

2. (a) $\int \frac{\sin x dx}{(1 - \cos x)^2}$ (b) $\int \frac{(\arctan x)^2 dx}{1 + x^2}$

(b) $\int \frac{(\arctan x)^2 dx}{1+x^2}$

3. (a) $\int \cos^3 x \, dx$ (b) $\int \frac{\cos x \, dx}{\sqrt[3]{\sin^2 x}}$ (c) $\int \cos^3 x \sin 2x \, dx$ (d) $\int \sec^2 x \tan x \, dx$

(b) $\int \frac{\cos x \, dx}{\sqrt[3]{\sin^2 x}}$

(c) $\int \cos^3 x \sin 2x dx$ (d) $\int \sec^2 x \tan x dx$

4. $\int \frac{dx}{\cos^2 x \sqrt{1 + \tan x}}$

5. $\int \frac{\sin 2x}{1 + \cos^2 x} dx$

6. $\int e^{\sin x} \cos x dx$

7. $\int (\tan^2 x + \tan^4 x) dx$

8. (a) $\int \tan^3 x \, dx$

(b) $\int \tan^4 x \, dx$

9. $\int \frac{1 - \sin x}{\cos x} dx$

10. $\int \frac{dx}{\cos^4 x}$

- | | | |
|--|---|--|
| 11. (a) $\int \frac{\sin^3 x}{\cos x} dx$ | (b) $\int \frac{\cos^3 x dx}{\sin^4 x}$ | 12. $\int \frac{\sin^3 \alpha}{\sqrt{\cos \alpha}} d\alpha$ |
| 13. $\int \frac{dx}{(\arcsin x)^3 \sqrt{1-x^2}}$ | 14. $\int \frac{dx}{1+\sin x}$ | 15. $\int \frac{1+\sin x}{1-\sin x} dx$ |
| 16. $\int \frac{dx}{\cos x}$ | 17. (a) $\int \frac{1-\sin x}{x+\cos x} dx$ | (b) $\int \frac{\cos 2x + x + 1}{x^2 + \sin 2x + 2x} dx$ |
| 18. $\int \cos^5 x dx$ | 19. $\int \frac{d\phi}{\sqrt{3} \cos \phi + \sin \phi}$ | 20. $\int \frac{dx}{\cot \frac{x}{2} \cdot \cot \frac{x}{3} \cdot \cot \frac{x}{6}}$ |
| 21. $\int \frac{\tan(\ln x) \tan\left(\ln \frac{x}{2}\right) \tan(\ln 2)}{x} dx$ | 22. $\int \left(\frac{x \cos x - \sin x}{x \sin x} \right) dx$ | 23. $\int \frac{\sin x}{\sin x + \cos x} dx$ |

(ii) Standard substitutions :

$$\int \frac{dx}{\sqrt{a^2 + x^2}} \text{ or } \int \sqrt{a^2 + x^2} dx ; \text{ put } x = a \tan \theta \text{ or } x = a \cot \theta$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} \text{ or } \int \sqrt{a^2 - x^2} dx ; \text{ put } x = a \sin \theta \text{ or } x = a \cos \theta$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} \text{ or } \int \sqrt{x^2 - a^2} dx ; \text{ put } x = a \sec \theta \text{ or } x = a \operatorname{cosec} \theta$$

$$\int \sqrt{\frac{a-x}{a+x}} dx ; \text{ put } x = a \cos 2\theta$$

$$\int \sqrt{\frac{x-\alpha}{\beta-x}} dx \text{ or } \int \sqrt{(x-\alpha)(\beta-x)} ; \text{ put } x = \alpha \cos^2 \theta + \beta \sin^2 \theta$$

$$\int \sqrt{\frac{x-\alpha}{x-\beta}} dx \text{ or } \int \sqrt{(x-\alpha)(x-\beta)} ; \text{ put } x = \alpha \sec^2 \theta - \beta \tan^2 \theta$$

$$\int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}} ; \text{ put } x - \alpha = t^2 \text{ or } x - \beta = t^2.$$

Formulae :

$$1. \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + c$$

$$\text{Sol. } \int \frac{dx}{a^2 + x^2} \quad \text{Put } x = a \tan \theta \quad dx = a \sec^2 \theta d\theta \Rightarrow \int \frac{a \sec^2 \theta d\theta}{a^2 + a^2 \tan^2 \theta} = \frac{1}{a} \int d\theta = \frac{1}{a} \theta = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$2. \int \frac{dx}{\sqrt{x^2 + a^2}} = \ln \left[x + \sqrt{x^2 + a^2} \right] + c$$

$$\text{Sol. } \int \frac{dx}{\sqrt{x^2 + a^2}} \quad \text{Put } x = a \tan \theta \quad dx = a \sec^2 \theta d\theta$$

$$\Rightarrow \int \frac{a \sec^2 \theta d\theta}{\sqrt{a^2 \tan^2 \theta + a^2}} = \int \sec \theta d\theta = \ln [\sec \theta + \tan \theta] = \ln \left[\sqrt{1 + \frac{x^2}{a^2}} + \frac{x}{a} \right] = \ln \left[\sqrt{x^2 + a^2} + x \right] + C$$

$$3. \int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{x+a}{x-a} \right| + C$$

Sol. $\int \frac{1}{(a-x)(a+x)} dx = \frac{1}{2a} \int \left(\frac{1}{a-x} + \frac{1}{a+x} \right) dx = \frac{1}{2a} [-\ln|x-a| + \ln|x+a|] + C$

$$4. \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + c$$

$$5. \int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left[x + \sqrt{x^2 - a^2} \right] + c$$

$$6. \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + c$$

Illustration 22 : Evaluate $\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} \cdot \frac{1}{x} dx$

Solution : Put $x = \cos^2 \theta \Rightarrow dx = -2 \sin \theta \cos \theta d\theta$

$$\begin{aligned} \Rightarrow I &= \int \sqrt{\frac{1-\cos \theta}{1+\cos \theta}} \cdot \frac{1}{\cos^2 \theta} (-2 \sin \theta \cos \theta) d\theta = -\int 2 \tan \frac{\theta}{2} \tan \theta d\theta \\ &= -4 \int \frac{\sin^2(\theta/2)}{\cos \theta} d\theta = -2 \int \frac{1-\cos \theta}{\cos \theta} d\theta = -2 \ln |\sec \theta + \tan \theta| + 2\theta + C \\ &= -2 \ln \left| \frac{1+\sqrt{1-x}}{\sqrt{x}} \right| + 2 \cos^{-1} \sqrt{x} + C \end{aligned}$$

Do yourself - 6 :

Evaluate the following :

1. $\int \frac{dx}{2x^2 + 9}$

2. $\int \frac{dx}{\sqrt{4-x^2}}$

3. $\int \frac{dx}{1+9x^2}$

4. $\int \frac{dx}{\sqrt{1-25x^2}}$

5. $\int \frac{dx}{\sqrt{9-16x^2}}$

6. $\int \frac{dx}{25+4x^2}$

7. $\int \frac{x^2}{9+16x^6} dx$

8. $\int \frac{dx}{x\sqrt{x^2+4}}$

9. $\int \frac{dx}{\sqrt{4-9x^2}}$

10. $\int \frac{dx}{\sqrt{4x-3-x^2}}$

11. $\int \frac{\cos \alpha d\alpha}{a^2 + \sin^2 \alpha}$

12. $\int \frac{xdx}{x^4+1}$

13. $\int \frac{xdx}{\sqrt{a^2-x^4}}$

14. $\int \frac{x^2 dx}{x^6+4}$

15. $\int \frac{x^3 dx}{\sqrt{1-x^8}}$

16. $\int \frac{e^x dx}{e^{2x}+4}$

17. $\int \frac{2^x dx}{\sqrt{1-4^x}}$

18. $\int \frac{1+x}{\sqrt{1-x^2}} dx$

19. $\int \frac{3x-1}{x^2+9} dx$

20. $\int \sqrt{\frac{1-x}{1+x}} dx$

21. $\int \sqrt{\frac{x-3}{2-x}} dx$

22. $\int \frac{x(1-x^2)}{1+x^4} dx$

23. $\int \frac{2x - \sqrt{\arcsin x}}{\sqrt{1-x^2}} dx$

24. $\int \frac{x + (\arccos 3x)^2}{\sqrt{1-9x^2}} dx$

25. $\int \frac{1+x-x^2}{\sqrt{(1-x^2)^3}} dx$

26. $\int \frac{dx}{(x-1)^2+4}$

27. $\int \frac{dx}{\sqrt{1-(2x+3)^2}}$

28. $\int \frac{2x+3}{\sqrt{1+x^2}} dx$

29. $\int \frac{dx}{x\sqrt{3-\ln^2 x}}$

30. $\int \frac{(2x^2-5)dx}{x^4-5x^2+6}$

31. $\frac{1}{12} \ln \left| \frac{2x-3}{2x+3} \right| + C$

32. $\frac{1}{2\sqrt{6}} \ln \left| \frac{\sqrt{2}+x\sqrt{3}}{\sqrt{2}-x\sqrt{3}} \right| + C$

Integral of the form :

(ii) $\int \frac{dx}{ax^2+bx+c}, \int \frac{dx}{\sqrt{ax^2+bx+c}}$

Express ax^2+bx+c in the form of perfect square & then apply the standard results.

(iii) $\int \frac{px+q}{ax^2+bx+c} dx, \int \frac{px+q}{\sqrt{ax^2+bx+c}} dx$

Express $px+q = \ell$ (differential coefficient of denominator) + m.

Illustration 23: $\int \frac{\sin x dx}{\cos^2 x - 36}$

Solution : put $\cos x = t \Rightarrow -\sin x dx = dt$

$$\int \frac{-dt}{t^2 - (6)^2} = -\frac{1}{2 \cdot 6} \ln \left| \frac{t-6}{t+6} \right| + C$$

$$= -\frac{1}{12} \ln \left| \frac{\cos x - 6}{\cos x + 6} \right| + C$$

Illustration 24: $\int \frac{dx}{3x^2+6x+15}$

Solution :

$$\Rightarrow \frac{1}{3} \int \frac{dx}{x^2+2x+5}$$

$$\Rightarrow \frac{1}{3} \int \frac{dx}{(x+1)^2+4} = \frac{1}{3} \int \frac{dx}{(x+1)^2+(2)^2}$$

$$= \frac{1}{6} \tan^{-1} \left(\frac{x+1}{2} \right) + C$$

Illustration 25: $\int \frac{dx}{\sqrt{x^2 - 5x}}$

Solution :

$$\int \frac{dx}{\sqrt{\left(x - \frac{5}{2}\right)^2 - \left(\frac{5}{2}\right)^2}} = \ln \left[\left(x - \frac{5}{2}\right) + \sqrt{\left(x - \frac{5}{2}\right)^2 - \left(\frac{5}{2}\right)^2} \right] + C$$

Illustration 26: $\int \frac{2x}{\sqrt{x^4 + 2x^2 + 4}} dx$

Solution :

$$\text{put } x^2 = t \Rightarrow 2x \, dx = dt$$
$$\int \frac{dt}{\sqrt{t^2 + 2t + 4}} = \int \frac{dt}{\sqrt{(t+1)^2 + (\sqrt{3})^2}}$$
$$= \ln \left| (t+1) + \sqrt{(t+1)^2 + (\sqrt{3})^2} \right| + C$$
$$t = x^2$$

Illustration 27: Evaluate $\int \frac{dx}{\sqrt{(x-a)(b-x)}}$

Solution : Put $x = a\cos^2\theta + b\sin^2\theta$, the given integral becomes

$$\begin{aligned} I &= \int \frac{2(b-a) \sin \theta \cos \theta d\theta}{\{(a \cos^2 \theta + b \sin^2 \theta - a)(b - a \cos^2 \theta - b \sin^2 \theta)\}^{\frac{1}{2}}} \\ &= \int \frac{2(b-a) \sin \theta \cos \theta d\theta}{(b-a) \sin \theta \cos \theta} = \left(\frac{b-a}{b-a} \right) \int 2d\theta = 2\theta + C = 2 \sin^{-1} \sqrt{\left(\frac{x-a}{b-a} \right)} + C \end{aligned}$$

Ans.

Illustration 28 : Evaluate $\int \frac{dx}{2x^2 + x - 1}$

Solution :

$$\begin{aligned}
 I &= \int \frac{dx}{2x^2 + x - 1} = \frac{1}{2} \int \frac{dx}{x^2 + \frac{x}{2} - \frac{1}{2}} = \frac{1}{2} \int \frac{dx}{x^2 + \frac{x}{2} + \frac{1}{16} - \frac{1}{16} - \frac{1}{2}} \\
 &= \frac{1}{2} \int \frac{dx}{(x + 1/4)^2 - 9/16} = \frac{1}{2} \int \frac{dx}{(x + 1/4)^2 - (3/4)^2} \\
 &= \frac{1}{2} \cdot \frac{1}{2(3/4)} \log \left| \frac{x + 1/4 - 3/4}{x + 1/4 + 3/4} \right| + C \quad \left\{ \text{using, } \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x - a}{x + a} \right| + C \right\} \\
 &= \frac{1}{3} \log \left| \frac{x - 1/2}{x + 1} \right| + C = \frac{1}{3} \log \left| \frac{2x - 1}{2(x + 1)} \right| + C
 \end{aligned}$$

Ans.

Illustration 29 : Evaluate $\int \frac{3x+2}{4x^2+4x+5} dx$

Solution : Express $3x+2 = \ell(\text{d.c. of } 4x^2+4x+5) + m$

$$\text{or, } 3x+2 = \ell(8x+4) + m$$

Comparing the coefficients, we get

$$8\ell = 3 \text{ and } 4\ell + m = 2 \Rightarrow \ell = 3/8 \text{ and } m = 2 - 4\ell = 1/2$$

$$\Rightarrow I = \frac{3}{8} \int \frac{8x+4}{4x^2+4x+5} dx + \frac{1}{2} \int \frac{dx}{4x^2+4x+5}$$

$$= \frac{3}{8} \log |4x^2+4x+5| + \frac{1}{8} \int \frac{dx}{x^2+x+\frac{5}{4}}$$

$$= \frac{3}{8} \log |4x^2+4x+5| + \frac{1}{8} \tan^{-1} \left(x + \frac{1}{2} \right) + C$$

Ans.

Do yourself - 7 :

1. $\int \frac{dx}{x^2+2x+3}$

2. $\int \frac{dx}{x-x^2-2.5}$

3. $\int \frac{dx}{4x^2+4x+5}$

4. $\int \frac{dx}{\sqrt{8+6x-9x^2}}$

5. $\int \frac{dx}{\sqrt{2-6x-9x^2}}$

6. $\int \frac{x dx}{x^4-3x^2+2}$

7. $\int \frac{dx}{x^4-x^2}$

8. $\int \frac{dx}{x^2+x+1}$

9. $\int \frac{dx}{\sqrt{x-x^2}}$

10. $\int \frac{dx}{x^2-7x+10}$

11. $\int \frac{x^2-x+1}{\sqrt{(x^2+1)^3}} dx$

12. $\int \frac{(x+2)dx}{x^2+2x+2}$

13. $\int \frac{(x-3)dx}{\sqrt{3-2x-x^2}}$

14. $\int \frac{(3x-1)dx}{\sqrt{x^2+2x+2}}$

15. $\int \frac{dx}{x^2+3x-10}$

16. $\int \frac{5x+4}{\sqrt{x^2+4x+1}} dx$

(b) **Integration by part :** $\int u \cdot v dx = u \int v dx - \int \left[\frac{du}{dx} \cdot \int v dx \right] dx$ where u & v are differentiable functions and are commonly designated as first & second function respectively.

Note : While using integration by parts, choose u & v such that

(i) $\int v dx$ & (ii) $\int \left[\frac{du}{dx} \cdot \int v dx \right] dx$ are simple to integrate.

This is generally obtained by choosing first function as the function which comes first in the word **ILATE**, where; I-Inverse function, L-Logarithmic function, A-Algebraic function, T-Trigonometric function & E-Exponential function.

$$\text{Let } I = \int \underbrace{f(x)}_I \cdot \underbrace{g(x)}_II \, dx = f(x) \cdot \int g(x) \, dx - \int (f'(x)) \left(\int g(x) \, dx \right) dx$$

$$= 1^{\text{st}} \text{ function} \times \text{integral of 2nd} - \int (\text{diff. coeff. of } I^{\text{st}}) \times (\text{integral of } 2^{\text{nd}}) dx$$

Proof : $\frac{d}{dx}[f(x) \cdot g(x)] = f(x) \cdot g'(x) + g(x) \cdot f'(x)$

$$\therefore \int \underbrace{f(x)}_I \cdot \underbrace{g'(x)}_II \, dx = f(x) \cdot g(x) - \int g(x) \cdot f'(x) \, dx$$

Note : In applying the above rule care has to be taken in the selection of the first function(I) and the second function (II). Normally we use the following methods :

- (i) If in the product of the two functions, one of the functions is not directly integrable (e.g. $\ln x$, $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$ etc.) then we take it as the first function and the remaining function is taken as the second function. e.g. in the integration of $\int x \tan^{-1} x \, dx$, $\tan^{-1} x$ is taken as the first function and x as the second function.
- (ii) If there is no other function, then unity is taken as the second function e.g. in the integration of $\int \tan^{-1} x \, dx$, $\tan^{-1} x$ is taken as the first function and 1 as the second function.
- (iii) If both of the functions are directly integrable then the first function is chosen in such a way that the derivative of the function thus obtained under integral sign is easily integrable. Usually we use the following preference order for the function (Inverse, Logarithmic, Algebraic, Trigonometric, Exponent). In the above stated order, the function on the left is always chosen as the first function. This rule is called as ILATE e.g. in the integration of $\int x \sin x \, dx$, x is taken as the first function and $\sin x$ is taken as the second function.

Illustration 30 : $\int e^{\ln x + x} \cdot dx$

Solution :

$$\begin{aligned}
 & x \int e^x \, dx - \int (1) \left(\int e^x \, dx \right) dx \\
 & x e^x - \int e^x \, dx \\
 & x e^x - e^x + C
 \end{aligned}$$

Illustration 31 : $\int \underbrace{x^4}_{\text{II}} \underbrace{\ln x}_{\text{I}} dx$

Solution : $\Rightarrow \ln x \int x^4 dx - \int \left(\frac{d(\ln x)}{dx} \int x^4 dx \right) dx$

$$\Rightarrow \ln x \cdot \frac{x^5}{5} - \int \frac{1}{x} \times \frac{x^5}{5} dx$$

$$\Rightarrow \frac{x^5}{5} \ln x - \frac{x^5}{25} + C$$

Illustration 32 : $\int x \cos 2x \, dx$

Solution : $\Rightarrow x \int \cos 2x \, dx - \int (1) \left(\int \cos 2x \, dx \right) dx$

$$\Rightarrow \frac{x \sin 2x}{2} - \int \frac{\sin 2x}{2} dx$$

$$\Rightarrow \frac{x \sin 2x}{2} + \frac{\cos 2x}{4} + C$$

Illustration 33 : $\int \ln(2x+3)^{(2x+3)} dx$

Solution : $\Rightarrow \int \underbrace{(2x+3)}_{\text{II}} \underbrace{\ln(2x+3)}_{\text{I}} dx$

$$\Rightarrow \ln(2x+3) \int (2x+3) dx - \int \frac{1}{(2x+3)} \left(\int (2x+3) dx \right) dx$$

$$\Rightarrow \ln(2x+3) \frac{(2x+3)^2}{4} - \int \frac{1}{2x+3} \times \frac{(2x+3)^2}{2} dx$$

$$\Rightarrow \ln(2x+3) \frac{(2x+3)^2}{4} - \frac{(2x+3)^2}{4} + C$$

Illustration 34 : $\int \underbrace{x^2}_{\text{I}} \underbrace{e^x}_{\text{II}} dx$

Solution : $x^2(e^x) - \int \underbrace{2x}_{\text{I}} \underbrace{e^x}_{\text{II}} dx$

$$x^2(e^x) - [2xe^x - \int 2e^x dx] = x^2e^x - 2xe^x + 2e^x + C$$

Illustration 35: $\int \sin^{-1} x \, dx$

Solution :

M-1 :

Let 1 is algebraic function

$$\int \underbrace{1}_{\text{II}} \cdot \underbrace{\sin^{-1} x}_{\text{I}} \, dx$$

$$\sin^{-1} x(x) - \int \left(\frac{1}{\sqrt{1-x^2}} \right) (x) \, dx$$

$$1 - x^2 = t$$

$$-2x \, dx = dt$$

$$x \sin^{-1}(x) + \frac{1}{2} \int \frac{dt}{\sqrt{t}}$$

$$x \sin^{-1} x + \frac{1}{2} \frac{\sqrt{t}}{\left(\frac{1}{2}\right)} + C$$

$$x \sin^{-1} x + \sqrt{1-x^2} + C$$

M-2 :

$$\sin^{-1} x = t$$

$$x = \sin t$$

$$dx = \cos t \, dt$$

$$\int \underbrace{t}_{\text{I}} \cdot \underbrace{\cos t}_{\text{II}} \, dt$$

$$t(\sin t) - \int (1) \sin t \, dt$$

$$t \sin t + \cos t + C$$

$$\sin^{-1} x(x) + \sqrt{1-x^2} + C$$

Illustration 36: $\int \ln x \, dx$

Solution :

M-1 :

$$\int \underbrace{1}_{\text{II}} \cdot \underbrace{\ln x}_{\text{I}} \, dx$$

$$\ln(x) \cdot (x) - \int \frac{1}{x} \cdot x \, dx$$

$$x \ln x - x + C$$

M-2 :

$$\int \ln x \, dx$$

Put, $\ln x = t$

$$x = e^t$$

$$dx = e^t \, dt$$

$$\int \underbrace{t}_{\text{I}} \cdot \underbrace{e^t}_{\text{II}} \, dt$$

Illustration 37: Evaluate : $\int \cos \sqrt{x} dx$

Solution : Consider $I = \int \cos \sqrt{x} dx$

$$\text{Let } \sqrt{x} = t \quad \text{then } \frac{1}{2\sqrt{x}} dx = dt$$

$$\text{i.e. } dx = 2\sqrt{x} dt \quad \text{or } dx = 2t dt$$

$$\text{so } I = \int \cos t \cdot 2t dt$$

Taking t as first function, integrate it by part

$$\Rightarrow I = 2 \left[t \int \cos t dt - \int \left\{ \frac{dt}{dt} \int \cos t dt \right\} dt \right]$$

$$I = 2 \left[t \sin t - \int 1 \cdot \sin t dt \right] = 2 [t \sin t + \cos t] + C$$

$$I = 2 [\sqrt{x} \sin \sqrt{x} + \cos \sqrt{x}] + C$$

Ans.

Illustration 38: Evaluate : $\int \frac{x}{1 + \sin x} dx$

Solution : Let $I = \int \frac{x}{1 + \sin x} dx = \int \frac{x(1 - \sin x)}{(1 + \sin x)(1 - \sin x)} dx$

$$= \int \frac{x(1 - \sin x)}{1 - \sin^2 x} dx = \int \frac{x(1 - \sin x)}{\cos^2 x} dx = \int x \sec^2 x dx - \int x \sec x \tan x dx$$

$$= \left[x \int \sec^2 x dx - \int \left\{ \frac{dx}{dx} \int \sec^2 x dx \right\} dx \right]$$

$$- \left[x \int \sec x \tan x dx - \int \left\{ \frac{dx}{dx} \int \sec x \tan x dx \right\} dx \right]$$

$$= [x \tan x - \int \tan x dx] - [x \sec x - \int \sec x dx]$$

$$= [x \tan x - \ln |\sec x|] - [x \sec x - \ln |\sec x + \tan x|] + C$$

$$= x(\tan x - \sec x) + \ln \left| \frac{(\sec x + \tan x)}{\sec x} \right| + C$$

$$= \frac{-x(1 - \sin x)}{\cos x} + \ln |1 + \sin x| + C$$

Ans.

Do yourself - 8 :

- | | | |
|--|---|--|
| 1. $\int x \ln x \, dx$ | 2. $\int x e^{-x} \, dx$ | 3. $\int x \sin^2 x \, dx$ |
| 4. $\int 2x^3 e^{x^2} \, dx$ | 5. $\int x e^x \, dx$ | 6. $\int x^3 \sin(x^2) \, dx$ |
| 7. $\int \frac{x^2 \tan^{-1} x}{1+x^2} \, dx$ | 8. $\int \frac{3x^2-1}{2x\sqrt{x}} \arctan x \, dx$ | 9. $\int x^n \ln x \, dx \, (n \neq -1)$ |
| 10. $\int \frac{\arcsin \sqrt{x}}{\sqrt{1-x}} \, dx$ | 11. $\int \frac{x \arctan x}{\sqrt{1+x^2}} \, dx$ | 12. $\int \frac{\ln^3 x}{x^2} \, dx$ |
| 13. $\int \frac{x^2 \, dx}{\sqrt{1-x^2}}$ | 14. $\int \arctan \sqrt{x} \, dx$ | 15. $\int \sqrt{e^x-1} \, dx$ |
| 16. $\int \ln(x + \sqrt{1+x^2}) \, dx$ | 17. $\int \sin(\ln x) \, dx$ | 18. $\int e^{ax} \cos nx \, dx$ |
| 19. $\int (\arctan x)^2 \, dx$ | | |

Two Classic Integrands :

(a) $\int e^x (f(x) + f'(x)) \, dx = e^x f(x) + c$

Proof:

Let $I = \int e^x [f(x) + f'(x)] \, dx$

$\Rightarrow I = \int e^x f(x) \, dx + \int e^x f'(x) \, dx$

Integrate the first integral on the RHS by parts taking e^x as the second function, we get

$I = e^x f(x) - \int e^x f'(x) \, dx + \int e^x f'(x) \, dx$

$\Rightarrow I = e^x f(x)$

Thus to evaluate the integrals of the type $\int e^x g(x) \, dx$, we first try to express $g(x)$ as the sum of the function and its derivative i.e., $g(x) = f(x) + f'(x)$ and then we use the result derived above.

Illustration 39: $\int e^x (\sin x + \cos x) \, dx$

Solution : $\int e^x (\sin x + \cos x) \, dx$
 $\quad \quad \quad \begin{matrix} f(x) & f'(x) \end{matrix}$
 $\Rightarrow e^x \sin x + C$

Illustration 40: $\int e^x \left(\frac{2x^2 + x^3}{x} \right) \, dx$

Solution : $\int e^x \left(\frac{2x + x^2}{f(x)} \right) \, dx$
 $\Rightarrow e^x \cdot x^2 + C$

Illustration 41 : $\int \frac{xe^x}{(1+x)^2} dx$

Solution :
$$\int e^x \left(\frac{x+1}{(x+1)^2} + \frac{(-1)}{(x+1)^2} \right) dx$$

$$\Rightarrow \int e^x \left(\underbrace{\frac{1}{x+1}}_{f(x)} + \underbrace{\frac{(-1)}{(x+1)^2}}_{f'(x)} \right) dx$$

$$= e^x \cdot \frac{1}{1+x} + C$$

Illustration 42 : $\int e^x \left(\frac{\sin 2x - 2}{1 - \cos 2x} \right) dx$

Solution :
$$\int e^x \left(\frac{2 \sin x \cos x - 2}{2 \sin^2 x} \right) dx$$

$$\int e^x \left(\frac{2 \sin x \cos x}{2 \sin^2 x} - \frac{2}{2 \sin^2 x} \right) dx$$

$$\int e^x \left(\underbrace{\cot x}_{f(x)} + \underbrace{(-\operatorname{cosec}^2 x)}_{f'(x)} \right) dx$$

$$\Rightarrow e^x \cot x + C$$

Illustration 43 : Evaluate $\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$

Solution :
$$\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx = \int e^x \frac{(1-2x+x^2)}{(1+x^2)^2} dx$$

$$= \int e^x \left(\frac{1}{(1+x^2)} - \frac{2x}{(1+x^2)^2} \right) dx = \frac{e^x}{1+x^2} + C$$

Ans.

Illustration 44 : The value of $\int e^x \left(\frac{x^4+2}{(1+x^2)^{5/2}} \right) dx$ is equal to -

(A) $\frac{e^x(x+1)}{(1+x^2)^{3/2}} + C$ (B) $\frac{e^x(1-x+x^2)}{(1+x^2)^{3/2}} + C$ (C) $\frac{e^x(1-x)}{(1+x^2)^{3/2}} + C$ (D) none of these

Solution : Let $I = \int e^x \left(\frac{x^4+2}{(1+x^2)^{5/2}} \right) dx = \int e^x \left(\frac{1}{(1+x^2)^{1/2}} + \frac{1-2x^2}{(1+x^2)^{5/2}} \right) dx$

$$= \int e^x \left(\frac{1}{(1+x^2)^{1/2}} - \frac{x}{(1+x^2)^{3/2}} + \frac{x}{(1+x^2)^{3/2}} + \frac{1-2x^2}{(1+x^2)^{5/2}} \right) dx$$

$$= \frac{e^x}{(1+x^2)^{1/2}} + \frac{xe^x}{(1+x^2)^{3/2}} + C = \frac{e^x \{1+x^2+x\}}{(1+x^2)^{3/2}} + C$$

Ans. (D)

Illustration 45 : Prove that $\int e^{g(x)} (g'(x) \cdot f(x) + f'(x)) dx = e^{g(x)} \cdot f(x)$.

Solution $I = \int e^{g(x)} (g'(x)f(x) + f'(x)) dx$
 $= \int e^{g(x)} g'(x) \cdot f(x) dx + \int e^{g(x)} f'(x) dx$
 Integrate the first integral on the R.H.S. by parts taking $e^{g(x)} \cdot g'(x)$ as the second function, we get
 $I = e^{g(x)} f(x) - \int f'(x) e^{g(x)} dx + \int e^{g(x)} \cdot f'(x) dx$
 $= e^{g(x)} \cdot f(x)$

(b) $\int (f(x) + xf'(x)) dx = xf(x) + c$

Proof : $= \int f(x) \cdot 1 dx + \int xf'(x) dx$
 $= \left[f(x) \cdot x - \int f'(x) \cdot x \right] + \int xf'(x) dx$
 $= xf(x) + c$

Illustration 46: $\int (\sin x + x \cos x) dx$

Solution : $\Rightarrow \int \underbrace{(\sin x)}_{f(x)} + x \underbrace{\cos x}_{f'(x)} dx$
 $\Rightarrow x \sin x + C$

Illustration 47: $\int \left(\ln(\ln x^2) + \frac{2}{\ln x^2} \right) dx$

Solution : $\int \underbrace{(\ln(\ln x^2))}_{f(x)} + \underbrace{\left(\frac{1}{\ln x^2} \times \frac{1}{x^2} \times 2x \right)}_{f'(x)} dx$
 $\Rightarrow x \ln(\ln x^2) + C$

Illustration 48: $\int e^{\tan^{-1}x} \left(\frac{1+x+x^2}{1+x^2} \right) dx$

Solution : $\int e^{\tan^{-1}x} \left(\frac{1+x^2}{1+x^2} + \frac{x}{1+x^2} \right) dx = \int e^{\tan^{-1}x} \left(1 + \frac{x}{1+x^2} \right) dx$
 $= \int \underbrace{e^{\tan^{-1}x}}_{f(x)} + x \cdot \underbrace{\frac{e^{\tan^{-1}x}}{1+x^2}}_{xf'(x)} dx = xe^{\tan^{-1}x} + C$

Illustration 49: Evaluate $\int \frac{x + \sin x}{1 + \cos x} dx$

Solution :
$$I = \int \frac{x + \sin x}{1 + \cos x} dx = \int \left(\frac{x + \sin x}{2 \cos^2 \frac{x}{2}} \right) dx = \int \left(x \frac{1}{2} \sec^2 \frac{x}{2} + \tan \frac{x}{2} \right) dx = x \tan \frac{x}{2} + C \quad \text{Ans.}$$

Integrals of the Type :

$$\int e^{ax} \sin bx \, dx, \int e^{ax} \cos bx \, dx$$

Results :

$$(i) \quad \int e^{ax} \cdot \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + c$$

$$(ii) \quad \int e^{ax} \cdot \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + c$$

Proof :

Method-I Let $I = \int e^{ax} \sin bx \, dx$

Integrate by parts taking e^{ax} as the first part, we get

$$I = \frac{-e^{ax} \cos bx}{b} - \int a e^{ax} \left(\frac{-\cos bx}{b} \right) dx$$

On integrating the second term by parts again, we get

$$I = -\frac{1}{b} e^{ax} \cos bx + \frac{a}{b} \left[\frac{1}{b} e^{ax} \sin bx - \int \frac{a}{b} e^{ax} \sin bx \, dx \right]$$

$$\Rightarrow I = -\frac{1}{b} e^{ax} \cos bx + \frac{a}{b^2} e^{ax} \sin bx - \frac{a^2}{b^2} I$$

$$\Rightarrow \left(1 + \frac{a^2}{b^2} \right) I = \frac{e^{ax}}{b^2} (a \sin bx - b \cos bx) + C$$

$$\Rightarrow I = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C$$

$$\text{Similarly we can show that } \int e^{ax} \cos bx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C$$

Illustration 50 : Evaluate $I = \int e^x \cdot \sin x \, dx = \sin x \cdot e^x - \int \cos x \cdot e^x \, dx$

Solution :

$$= \sin x e^x - \left[\cos x e^x - \int (-\sin x) e^x dx \right]$$

$$= \sin x e^x - \cos x e^x - I$$

$$\Rightarrow 2I = \sin x e^x - \cos x e^x$$

$$\Rightarrow I = \frac{1}{2} (\sin x - \cos x) e^x + c$$

Method-II Assume $\int e^{2x} \cos 3x \, dx = e^{2x} (A \cos 3x + B \sin 3x)$ and then differentiate both sides

Illustration 51 : $\int e^{2x} \cos 3x \, dx = e^{2x} \{A \cos 3x + B \sin 3x\} + C$ find A & B.

Solution : differentiate both sides

$$e^{2x} \cos 3x = e^{2x} (-3A \sin 3x + 3B \cos 3x) + 2e^{2x} (A \cos 3x + B \sin 3x)$$

$$= e^{2x} [(2A + 3B) \cos 3x + (2B - 3A) \sin 3x]$$

$$\Rightarrow \begin{aligned} 2A + 3B &= 1 \\ 2B - 3A &= 0 \end{aligned}$$

$$\Rightarrow A = \frac{2}{13} \text{ and } B = \frac{3}{13}$$

Ans.

Do yourself - 9 :

- $\int e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right) dx$
- $\int x e^{x^2} (\sin x^2 + \cos x^2) dx$
- $\int (\tan(e^x) + x e^x \sec^2(e^x)) dx$
- $\int \frac{(x + \sin x) dx}{1 + \cos x}$
- $\int (\ln x + 1) dx$
- $\int e^{\sin x} \frac{x \cos^3 x - \sin x}{\cos^2 x} dx$
- $\int \frac{e^{\cos x} (x \sin^3 x + \cos x)}{\sin^2 x} dx$
- The value of $\int \left\{ \frac{(\log x - 1)}{1 + (\log x)^2} \right\}^2 dx$ is equal to :
 (A) $\frac{\log x}{(\log x)^2 + 1} + C$
 (B) $\frac{x}{x^2 + 1} + C$
 (C) $\frac{x e^x}{1 + x^2} + C$
 (D) $\frac{x}{(\log x)^2 + 1} + C$
- The value of $\int e^{\tan \theta} (\sec \theta - \sin \theta) d\theta$ is equal to :
 (A) $-e^{\tan \theta} \sin \theta + C$
 (B) $e^{\tan \theta} \sin \theta + C$
 (C) $-e^{\tan \theta} \sec \theta + C$
 (D) $e^{\tan \theta} \cos \theta + C$

10. $\int e^{-2x} \sin 3x \, dx =$

(A) $\frac{1}{13} e^{-2x} [\sin 3x + \cos 3x] + c$

(B) $-\frac{1}{13} e^{-2x} [\sin 3x + \cos 3x] + c$

(C) $\frac{1}{13} e^{-2x} [2 \sin 3x + 3 \cos 3x] + c$

(D) $-\frac{1}{13} e^{-2x} [2 \sin 3x + 3 \cos 3x] + c$

11. $\int e^x (\log(\sin x) + \cot x) \, dx =$

(A) $e^x \cot x + c$

(B) $e^x \log(\sin x) + c$

(C) $e^x \log(\sin x) + \tan x + c$

(D) $e^x + \sin x + c$

12. $\int e^x (1 + \tan x) \sec x \, dx =$

(A) $e^x \cot x$

(B) $e^x \tan x$

(C) $e^x \sec x$

(D) $e^x \cos x$

13. $\int e^x \left[\sin^{-1} \frac{x}{a} + \frac{1}{\sqrt{a^2 - x^2}} \right] dx =$

(A) $\frac{1}{a} e^x \sin^{-1} \frac{x}{a} + c$

(B) $a e^x \sin^{-1} \frac{x}{a} + c$

(C) $e^x \sin^{-1} \frac{x}{a} + c$

(D) $\frac{e^x}{\sqrt{a^2 - x^2}} + c$

14. If $u = \int e^{ax} \cos bx \, dx$ and $v = \int e^{ax} \sin bx \, dx$, then $(a^2 + b^2)(u^2 + v^2) =$

(A) $2e^{ax}$

(B) $(a^2 + b^2)e^{2ax}$

(C) e^{2ax}

(D) $(a^2 - b^2)e^{2ax}$

15. $\int e^{x/2} \sin \left(\frac{x}{2} + \frac{\pi}{4} \right) dx =$

(A) $e^{x/2} \cos \frac{x}{2} + c$

(B) $\sqrt{2} e^{x/2} \cos \frac{x}{2} + c$

(C) $e^{x/2} \sin \frac{x}{2} + c$

(D) $\sqrt{2} e^{x/2} \sin \frac{x}{2} + c$

(c) **Integration of trigonometric functions :**

(i) $\int \sin^m x \cos^n x \, dx$

Case-I : When m & $n \in$ natural numbers.

- * If one of them is odd, then substitute for the term of even power.
- * If both are odd, substitute either of the term.
- * If both are even, use trigonometric identities to convert integrand into cosines of multiple angles.

Case-II : $m + n$ is a negative even integer.

- * In this case the best substitution is $\tan x = t$.

Illustration 52 : Evaluate $\int \sin^3 x \cos^5 x \, dx$

Solution : Put $\cos x = t$; $-\sin x \, dx = dt$.

so that $I = -\int (1 - t^2) \cdot t^5 \, dt$

$$= \int (t^7 - t^5) \, dt = \frac{t^8}{8} - \frac{t^6}{6} = \frac{\cos^8 x}{8} - \frac{\cos^6 x}{6} + C$$

Alternate :

Put $\sin x = t$; $\cos x \, dx = dt$

so that $I = \int t^3 (1 - t^2)^2 \, dt = \int (t^3 - 2t^5 + t^7) \, dt$

$$= \frac{\sin^4 x}{4} - \frac{2\sin^6 x}{6} + \frac{\sin^8 x}{8} + C$$

Note : This problem can also be handled by successive reduction or by trigonometric identities.

Illustration 53 : Evaluate $\int \sin^2 x \cos^4 x \, dx$

Solution :

$$\begin{aligned} \int \sin^2 x \cos^4 x \, dx &= \int \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{\cos 2x + 1}{2} \right)^2 \, dx \\ &= \int \frac{1}{8} (1 - \cos 2x) (\cos^2 2x + 2\cos 2x + 1) \, dx \\ &= \frac{1}{8} \int (\cos^2 2x + 2\cos 2x + 1 - \cos^3 2x - 2\cos^2 2x - \cos 2x) \, dx \\ &= \frac{1}{8} \int (-\cos^3 2x - \cos^2 2x + \cos 2x + 1) \, dx \\ &= -\frac{1}{8} \int \left(\frac{\cos 6x + 3\cos 2x}{4} + \frac{1 + \cos 4x}{2} - \cos 2x - 1 \right) \, dx \\ &= -\frac{1}{32} \left[\frac{\sin 6x}{6} + \frac{3\sin 2x}{2} \right] - \frac{1}{16} x - \frac{\sin 4x}{64} + \frac{\sin 2x}{16} + \frac{x}{8} + C \\ &= -\frac{\sin 6x}{192} - \frac{\sin 4x}{64} + \frac{1}{64} \sin 2x + \frac{x}{16} + C \end{aligned}$$

Illustration 54 : Evaluate $\int \frac{\sqrt{\sin x}}{\cos^{9/2} x} \, dx$

Solution : Let $I = \int \frac{\sin^{1/2} x}{\cos^{9/2} x} \, dx = \int \frac{dx}{\sin^{-1/2} x \cos^{9/2} x}$

Here $m + n = \frac{1}{2} - \frac{9}{2} = -4$ (negative even integer).

Divide Numerator & Denominator by $\cos^4 x$.

$$I = \int \sqrt{\tan x} \sec^4 x \, dx = \int \sqrt{\tan x} (1 + \tan^2 x) \sec^2 x \, dx$$

$$= \int \sqrt{t} (1 + t^2) \, dt \quad (\text{using } \tan x = t)$$

$$= \frac{2}{3} t^{3/2} + \frac{2}{7} t^{7/2} + C = \frac{2}{3} \tan^{3/2} x + \frac{2}{7} \tan^{7/2} x + C$$

(iii) $\int \frac{dx}{a + b \sin x}$ OR $\int \frac{dx}{a + b \cos x}$ OR $\int \frac{dx}{a + b \sin x + c \cos x}$

& put $\tan \frac{x}{2} = t$

In this case $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $x = 2\tan^{-1}t$; $dx = \frac{2dt}{1+t^2}$

Illustration 57: $\int \frac{dx}{3+4 \sin 2 x}$

Solution : $\int \frac{dx}{3 + \frac{4.2 \tan x}{1 + \tan^2 x}}$

$$= \int \frac{(1 + \tan^2 x) dx}{3 + 3 \tan^2 x + 8 \tan x}$$

$$\begin{aligned} \text{Let } \tan x &= t \\ \sec^2 x dx &= dt \end{aligned}$$

$$I = \int \frac{dt}{3t^2 + 8t + 3}$$

$$= \frac{1}{3} \int \frac{dt}{\left(t + \frac{4}{3}\right)^2 - \frac{7}{9}}$$

$$= \frac{1}{3} \cdot \frac{3}{2\sqrt{7}} \ell n \left| \frac{t + \frac{4}{3} - \frac{\sqrt{7}}{3}}{t + \frac{4}{3} + \frac{\sqrt{7}}{3}} \right| + c$$

$$= \frac{1}{2\sqrt{7}} \ln \left| \frac{\tan x + \frac{4}{3} - \frac{\sqrt{7}}{3}}{\tan x + \frac{4}{3} + \frac{\sqrt{7}}{3}} \right| + c$$

Illustration 58: Evaluate: $\int \frac{dx}{3 \sin x + 4 \cos x}$

Solution :

$$I = \int \frac{dx}{3 \sin x + 4 \cos x} = \int \frac{dx}{3 \left\{ \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right\} + 4 \left\{ \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right\}} = \int \frac{\sec^2 \frac{x}{2} dx}{4 + 6 \tan \frac{x}{2} - 4 \tan^2 \frac{x}{2}}$$

$$\text{let } \tan \frac{x}{2} = t, \quad \therefore \frac{1}{2} \sec^2 \frac{x}{2} dx = dt$$

$$\text{so } I = \int \frac{2dt}{4+6t-4t^2} = \frac{1}{2} \int \frac{dt}{1-\left(t^2 - \frac{3}{2}t\right)} = \frac{1}{2} \int \frac{dt}{\frac{25}{16} - \left(t - \frac{3}{4}\right)^2}$$

$$= \frac{1}{2} \cdot \frac{1}{2\left(\frac{5}{4}\right)} \ln \left| \frac{\frac{5}{4} + \left(t - \frac{3}{4}\right)}{\frac{5}{4} - \left(t - \frac{3}{4}\right)} \right| + C = \frac{1}{5} \ln \left| \frac{1 + 2 \tan \frac{x}{2}}{4 - 2 \tan \frac{x}{2}} \right| + C$$

Ans.

Integral of the form :

$$(iv) \int \frac{a \cos x + b \sin x + c}{p \cos x + q \sin x + r} dx$$

Express Numerator (N^r) = $\ell(D^r) + m \frac{d}{dx}(D^r) + n$ & proceed.

Illustration 59 : Evaluate : $\int \frac{2+3 \cos \theta}{\sin \theta + 2 \cos \theta + 3} d\theta$

Solution :

Write the Numerator = $\ell(\text{denominator}) + m(\text{d.c. of denominator}) + n$

$$\Rightarrow 2 + 3 \cos \theta = \ell(\sin \theta + 2 \cos \theta + 3) + m(\cos \theta - 2 \sin \theta) + n.$$

Comparing the coefficients of $\sin \theta$, $\cos \theta$ and constant terms,

$$\text{we get } 3\ell + n = 2, \quad 2\ell + m = 3, \quad \ell - 2m = 0 \Rightarrow \ell = 6/5, m = 3/5 \text{ and } n = -8/5$$

$$\text{Hence } I = \int \frac{6}{5} d\theta + \frac{3}{5} \int \frac{\cos \theta - 2 \sin \theta}{\sin \theta + 2 \cos \theta + 3} d\theta - \frac{8}{5} \int \frac{d\theta}{\sin \theta + 2 \cos \theta + 3}$$

$$= \frac{6}{5} \theta + \frac{3}{5} \ln |\sin \theta + 2 \cos \theta + 3| - \frac{8}{5} I_3 \text{ where } I_3 = \int \frac{d\theta}{\sin \theta + 2 \cos \theta + 3}$$

$$\text{In } I_3, \text{ put } \tan \frac{\theta}{2} = t \Rightarrow \sec^2 \frac{\theta}{2} d\theta = 2dt$$

$$I_3 = 2 \int \frac{dt}{t^2 + 2t + 5} = 2 \int \frac{dt}{(t+1)^2 + 2^2} = 2 \cdot \frac{1}{2} \tan^{-1} \left(\frac{t+1}{2} \right) = \tan^{-1} \left(\frac{\tan \theta / 2 + 1}{2} \right)$$

$$\text{Hence } I = \frac{6\theta}{5} + \frac{3}{5} \ln |\sin \theta + 2 \cos \theta + 3| - \frac{8}{5} \tan^{-1} \left(\frac{\tan \theta / 2 + 1}{2} \right) + C$$

Ans.

Do yourself - 11 :

1. $\int \frac{dx}{1+4\sin^2 x}$

2. $\int \frac{dx}{4 - 3 \cos^2 x + 5 \sin^2 x}$

3. $\int \frac{dx}{3 \sin^2 x + \sin x \cos x + 1}$

4. $\int \frac{dx}{3 + \sin x}$

5. $\int \frac{dx}{5 - 4 \sin x + 3 \cos x}$

6. $\int \frac{dx}{1 + 4 \sin x + 3 \cos x}$

7. $\int \frac{3 \sin x + 2 \cos x}{3 \cos x + 2 \sin x} dx$

8. $\int \frac{\cos 2x \, dx}{1 + \sin x \cos x}$

9. $\int \frac{\sin 2x dx}{\cos^4 x + \sin^4 x}$

10. $\int \frac{dx}{(\sin x + 2 \sec x)^2}$

(d) Integration of rational function :

(i) Rational function is defined as the ratio of two polynomials in the form $\frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ are polynomials in x and $Q(x) \neq 0$. If the degree of $P(x)$ is less than the degree of $Q(x)$, then the rational function is called proper, otherwise, it is called improper. The improper rational function can be reduced to the proper rational functions by long division process. Thus, if $\frac{P(x)}{Q(x)}$ is improper, then $\frac{P(x)}{Q(x)} = T(x) + \frac{P_1(x)}{Q(x)}$, where $T(x)$ is a polynomial in x and $\frac{P_1(x)}{Q(x)}$ is proper rational function. It is always possible to write the integrand as a sum of simpler rational functions by a method called partial fraction decomposition. After this, the integration can be carried out easily using the already known methods.

S. No.	Form of the rational function	Form of the partial fraction
1.	$\frac{px^2 + qx + r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$
2.	$\frac{px^2 + qx + r}{(x-a)^2(x-b)}$	$\frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$
3.	$\frac{px^2 + qx + r}{(x-a)(x^2 + bx + c)}$ where $x^2 + bx + c$ cannot be factorised further	$\frac{A}{x-a} + \frac{Bx + C}{x^2 + bx + c}$

Illustration 63 : Evaluate : $\int \frac{1}{x^4 + 5x^2 + 1} dx$

Solution :

$$\begin{aligned} I &= \frac{1}{2} \int \frac{2}{x^4 + 5x^2 + 1} dx \\ \Rightarrow I &= \frac{1}{2} \int \frac{1+x^2}{x^4 + 5x^2 + 1} dx + \frac{1}{2} \int \frac{1-x^2}{x^4 + 5x^2 + 1} dx \\ &= \frac{1}{2} \int \frac{1+1/x^2}{x^2 + 5 + 1/x^2} dx - \frac{1}{2} \int \frac{1-1/x^2}{x^2 + 5 + 1/x^2} dx \\ &\quad \{\text{dividing } N^r \text{ and } D^r \text{ by } x^2\} \\ &= \frac{1}{2} \int \frac{(1+1/x^2)}{(x-1/x)^2 + 7} dx - \frac{1}{2} \int \frac{(1-1/x^2)}{(x+1/x)^2 + 3} dx \\ &= \frac{1}{2} \int \frac{dt}{t^2 + (\sqrt{7})^2} - \frac{1}{2} \int \frac{du}{u^2 + (\sqrt{3})^2} \end{aligned}$$

where $t = x - \frac{1}{x}$ and $u = x + \frac{1}{x}$

$$\begin{aligned} I &= \frac{1}{2} \cdot \frac{1}{\sqrt{7}} \left(\tan^{-1} \frac{t}{\sqrt{7}} \right) - \frac{1}{2} \cdot \frac{1}{\sqrt{3}} \left(\tan^{-1} \frac{u}{\sqrt{3}} \right) + C \\ &= \frac{1}{2} \left[\frac{1}{\sqrt{7}} \tan^{-1} \left(\frac{x-1/x}{\sqrt{7}} \right) - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x+1/x}{\sqrt{3}} \right) \right] + C \end{aligned}$$

Ans.

Do yourself - 13 :

1. $\int \frac{x^2 + 1}{x^4 - x^2 + 1} dx$

2. $\int \frac{1}{1+x^4} dx$

3. $\int \frac{(e^{3x} + e^x) dx}{e^{4x} - e^{2x} + 1}$

4. $\int \frac{\sin^3 x \cdot dx}{(\cos^4 x + 3 \cos^2 x + 1) \tan^{-1}(\sec x + \cos x)}$

5. $\int \frac{(x^2 - 1) dx}{x \sqrt{x^4 + 3x^2 + 1}}$

(e) Integration of Irrational functions :

(i) $\int \frac{dx}{(ax+b)\sqrt{px+q}}$ & $\int \frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$; put $px+q = t^2$

(ii) $\int \frac{dx}{(ax+b)\sqrt{px^2+qx+r}}$, put $ax+b = \frac{1}{t}$; $\int \frac{dx}{(ax^2+bx+c)\sqrt{px^2+q}}$, put $x = \frac{1}{t}$

Illustration 64: Evaluate $\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}}.dx$

Solution : Let, $I = \int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}}.dx$ Put $x+1 = t^2 \Rightarrow dx = 2tdt$

$$\begin{aligned} \therefore I &= \int \frac{(t^2 - 1) + 2}{\{(t^2 - 1)^2 + 3(t^2 - 1) + 3\}\sqrt{t^2}} \cdot (2t)dt = 2 \int \frac{t^2 + 1}{t^4 + t^2 + 1} dt = 2 \int \frac{1 + 1/t^2}{t^2 + 1 + 1/t^2} dt \\ &= 2 \int \frac{1 + 1/t^2}{(t - 1/t)^2 + (\sqrt{3})^2} \cdot dt = 2 \int \frac{du}{u^2 + (\sqrt{3})^2} \quad \left\{ \text{where } u = t - \frac{1}{t} \right\} \\ &= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{u}{\sqrt{3}} \right) + C = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t^2 - 1}{\sqrt{3}t} \right) + C = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3}(x+1)} \right) + C \quad \text{Ans.} \end{aligned}$$

Illustration 65: Evaluate $\int \frac{dx}{(x-1)\sqrt{x^2+x+1}}$

Solution : Let, $I = \int \frac{dx}{(x-1)\sqrt{x^2+x+1}}$ put $x-1 = \frac{1}{t} \Rightarrow dx = -1/t^2 dt$

$$\begin{aligned} I &= \int \frac{-1/t^2 dt}{1/t \sqrt{\left(\frac{1}{t}+1\right)^2 + \left(\frac{1}{t}+1\right)+1}} = - \int \frac{dt}{\sqrt{3t^2+3t+1}} \\ &= -\frac{1}{\sqrt{3}} \int \frac{dt}{\sqrt{\left(t+\frac{1}{2}\right)^2 + 1/12}} = -\frac{1}{\sqrt{3}} \log \left| \left(t+\frac{1}{2}\right) + \sqrt{\left(t+\frac{1}{2}\right)^2 + 1/12} \right| + C \\ &= -\frac{1}{\sqrt{3}} \log \left| \left(\frac{1}{x-1} + \frac{1}{2}\right) + \sqrt{\frac{12\left(\frac{1}{x-1} + \frac{1}{2}\right)^2 + 1}{12}} \right| + C \end{aligned}$$

Illustration 66: Evaluate: $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

Solution : Let, $I = \int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$ Put $x = \frac{1}{t}$, So that $dx = -\frac{1}{t^2} dt$

$$\therefore I = \int \frac{-1/t^2 dt}{(1+1/t^2)\sqrt{1-1/t^2}} = -\int \frac{tdt}{(t^2+1)\sqrt{t^2-1}}$$

Again let, $t^2 = u$. So that $2t \, dt = du$.

$$= \frac{-1}{2} \int \frac{du}{(u+1)\sqrt{u-1}} \text{ which reduces to the form } \int \frac{dx}{P\sqrt{Q}} \text{ where both P and Q are linear}$$

so that we put $u - 1 = z^2$ so that $du = 2z \, dz$

$$\therefore I = -\frac{1}{2} \int \frac{2zdz}{(z^2+1+1)\sqrt{z^2}} = -\int \frac{dz}{(z^2+2)}$$

$$I = -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{z}{\sqrt{2}} \right) + C$$

$$I = -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{u-1}}{\sqrt{2}} \right) + C = -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{t^2-1}}{\sqrt{2}} \right) + C$$

$$= -\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2}x} \right) + C$$

Ans.

Do yourself - 14 :

1. $\int \frac{x}{(x-3)\sqrt{x+1}} dx$

2. $\int \frac{dx}{x^2\sqrt{1+x^2}}$

3. $\int \frac{1}{(x+1)\sqrt{x+2}} dx$

4. $\int \frac{1}{(x^2-4)\sqrt{x+1}} dx$

5. $\int \frac{dx}{x\sqrt{1+x^2}}$

6. $\int \frac{dx}{x^2\sqrt{x^2-9}}$

7. To find the value of $\int \frac{dx}{x\sqrt{2ax-x^2}}$, the suitable substitution is

(A) $x = a \cos t$

(B) $x = 2a \cos t$

(C) $x = 2a t$

(D) $x = 2a \sin^2 t$

8. $\int \frac{1}{x^2\sqrt{1+x^2}} dx =$

(A) $-\frac{\sqrt{1+x^2}}{x} + c$

(B) $\frac{\sqrt{1+x^2}}{x} + c$

(C) $-\frac{\sqrt{1-x^2}}{x} + c$

(D) $-\frac{\sqrt{x^2-1}}{x} + c$

9. If $\int \frac{1}{(1+x)\sqrt{x}} dx = f(x) + A$, where A is any arbitrary constant, then the function f(x) is

(A) $2 \tan^{-1} x$

(B) $2 \tan^{-1} \sqrt{x}$

(C) $2 \cot^{-1} \sqrt{x}$

(D) $\log_e(1+x)$

10. $\int \frac{dx}{(x^2-1)\sqrt{x+1}}$

11. $\int \frac{dx}{(1-x^2)\sqrt{2+x}}$

12. $\int \frac{dx}{(2x+3)\sqrt{x^2+3x}}$

Integrals of Some More Types

Here, we discuss some special types of standard integrals based on the technique of integration by parts :

(i) $\int \sqrt{x^2-a^2} dx$

(ii) $\int \sqrt{x^2+a^2} dx$

(iii) $\int \sqrt{a^2-x^2} dx$

(i) Let $I = \int \sqrt{x^2-a^2} dx$

Taking constant function 1 as the second function and integrating by parts, we have

$$I = x\sqrt{x^2-a^2} - \int \frac{1}{2} \frac{2x}{\sqrt{x^2-a^2}} dx$$

$$= x\sqrt{x^2-a^2} - \int \frac{x^2}{\sqrt{x^2-a^2}} dx$$

$$\begin{aligned}
 &= x\sqrt{x^2 - a^2} - \int \frac{x^2 - a^2 + a^2}{\sqrt{x^2 - a^2}} dx \\
 &= x\sqrt{x^2 - a^2} - \int \sqrt{x^2 - a^2} dx - a^2 \int \frac{dx}{\sqrt{x^2 - a^2}} \\
 &= x\sqrt{x^2 - a^2} - I - a^2 \int \frac{dx}{\sqrt{x^2 - a^2}}
 \end{aligned}$$

$$\text{or } 2I = x\sqrt{x^2 - a^2} - a^2 \int \frac{dx}{\sqrt{x^2 - a^2}}$$

$$\text{or } I = \int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log|x + \sqrt{x^2 - a^2}| + C$$

Similarly, integrating other two integrals, by parts, taking constant function 1 as the second function, we get

$$(ii) \quad \int \sqrt{x^2 + a^2} dx = \frac{1}{2} x \sqrt{x^2 + a^2} + \frac{a^2}{2} \log|x + \sqrt{x^2 + a^2}| + C$$

$$(iii) \quad \int \sqrt{a^2 - x^2} dx = \frac{1}{2} x \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$$

Alternatively, integrals (i), (ii) and (iii) can also be found by making trigonometric substitution $x = a \sec \theta$ in (i), $x = a \tan \theta$ in (ii) and $x = a \sin \theta$ in (iii) respectively.

Illustration 67: Find $\int \sqrt{x^2 + 2x + 5} dx$

Solution :

Note that

$$\int \sqrt{x^2 + 2x + 5} dx = \int \sqrt{(x+1)^2 + 4} dx$$

Put $x + 1 = y$, so that $dx = dy$. Then

$$\begin{aligned}
 \int \sqrt{x^2 + 2x + 5} dx &= \int \sqrt{y^2 + 2^2} dy \\
 &= \frac{1}{2} y \sqrt{y^2 + 4} + \frac{4}{2} \log|y + \sqrt{y^2 + 4}| + C \\
 &= \frac{1}{2} (x+1) \sqrt{x^2 + 2x + 5} + 2 \log|x + 1 + \sqrt{x^2 + 2x + 5}| + C
 \end{aligned}$$

Illustration 68: Find $\int \sqrt{3 - 2x - x^2} dx$

Solution :

$$\text{Note that } \int \sqrt{3 - 2x - x^2} dx = \int \sqrt{4 - (x+1)^2} dx$$

Put $x + 1 = y$ so that $dx = dy$

$$\begin{aligned}
 \text{Thus } \int \sqrt{3 - 2x - x^2} dx &= \int \sqrt{4 - y^2} dy \\
 &= \frac{1}{2} y \sqrt{4 - y^2} + \frac{4}{2} \sin^{-1} \frac{y}{2} + C \\
 &= \frac{1}{2} (x+1) \sqrt{3 - 2x - x^2} + 2 \sin^{-1} \left(\frac{x+1}{2} \right) + C
 \end{aligned}$$

Do yourself - 15 :

Integrate the functions :

- | | | |
|---------------------------|----------------------|-------------------------|
| 1. $\sqrt{1-4x^2}$ | 2. $\sqrt{x^2+4x+6}$ | 3. $\sqrt{1-4x-x^2}$ |
| 4. $\sqrt{1+3x-x^2}$ | 5. $x\sqrt{x+x^2}$ | 6. $(x+1)\sqrt{2x^2+3}$ |
| 7. $(x+3)\sqrt{3-4x-x^2}$ | | |

(f) Manipulating integrands :

(i) $\int \frac{dx}{x(x^n+1)}$, $n \in \mathbb{N}$, take x^n common & put $1+x^{-n}=t$.

(ii) $\int \frac{dx}{x^2(x^n+1)^{\frac{(n-1)}{n}}}$, $n \in \mathbb{N}$, take x^n common & put $1+x^{-n}=t^n$

(iii) $\int \frac{dx}{x^n(1+x^n)^{1/n}}$, take x^n common and put $1+x^{-n}=t^n$.

Illustration 69 : Evaluate : $\int \frac{dx}{x^n(1+x^n)^{1/n}}$

Solution :

$$\text{Let } I = \int \frac{dx}{x^n(1+x^n)^{1/n}} = \int \frac{dx}{x^{n+1} \left(1 + \frac{1}{x^n}\right)^{1/n}}$$

Put $1 + \frac{1}{x^n} = t^n$, then $\frac{1}{x^{n+1}} dx = -t^{n-1} dt$

$$I = - \int \frac{t^{n-1} dt}{t} = - \int t^{n-2} dt = -\frac{t^{n-1}}{n-1} + C = \frac{-1}{n-1} \left(1 + \frac{1}{x^n}\right)^{\frac{n-1}{n}} + C$$

Ans.**Do yourself - 16 :**

- | | | |
|---|--|--|
| 1. $\int \frac{dx}{x(x^2+1)}$ | 2. $\int \frac{dx}{x^2(x^3+1)^{2/3}}$ | 3. $\int \frac{dx}{x^3(x^3+1)^{1/3}}$ |
| 4. If $f(x) = \int \frac{5x^8+7x^6}{(x^2+1+2x^7)^2} dx$, ($x \geq 0$) and $f(0) = 0$, then the value of $f(1)$ is : | | |
| (A) $-\frac{1}{2}$ | (B) $\frac{1}{2}$ | (C) $-\frac{1}{4}$ |
| (D) $\frac{1}{4}$ | | |
| 5. $\int \frac{1+x^4}{(1-x^4)^{3/2}} dx$ | 6. $\int \frac{2x-4}{x^2\sqrt{x^2-2x+2}} dx$ | 7. $\int \frac{dx}{x(x^{100}+1)}$ |
| 8. $\int \frac{dx}{x^2(x^5+1)^{4/5}}$ | 9. $\int \frac{x \sin \alpha + 2}{(x^2+x \sin \alpha + 1)^{3/2}} dx$ | 10. $\int \frac{dx}{x^{10}(10+x^{10})^{1/10}}$ |

Miscellaneous Illustrations :

Illustration 70: Evaluate $\int \frac{\cos^4 x dx}{\sin^3 x \{\sin^5 x + \cos^5 x\}^{\frac{3}{5}}}$

Solution :
$$I = \int \frac{\cos^4 x}{\sin^3 x \{\sin^5 x + \cos^5 x\}^{\frac{3}{5}}} dx = \int \frac{\cos^4 x}{\sin^6 x \{1 + \cot^5 x\}^{\frac{3}{5}}} dx = \int \frac{\cot^4 x \operatorname{cosec}^2 x dx}{(1 + \cot^5 x)^{3/5}}$$

Put $1 + \cot^5 x = t$

$$5\cot^4 x \operatorname{cosec}^2 x dx = -dt$$

$$= -\frac{1}{5} \int \frac{dt}{t^{3/5}} = -\frac{1}{2} t^{2/5} + C = -\frac{1}{2} (1 + \cot^5 x)^{2/5} + C$$

Ans.

Illustration 71: $\int \frac{dx}{\cos^6 x + \sin^6 x}$ is equal to -

- (A) $\ell n|\tan x - \cot x| + C$ (B) $\ell n|\cot x - \tan x| + C$
(C) $\tan^{-1}(\tan x - \cot x) + C$ (D) $\tan^{-1}(-2\cot 2x) + C$

Solution : Let $I = \int \frac{dx}{\cos^6 x + \sin^6 x} = \int \frac{\sec^6 x}{1 + \tan^6 x} dx = \int \frac{(1 + \tan^2 x)^2 \sec^2 x dx}{1 + \tan^6 x}$

If $\tan x = p$, then $\sec^2 x \, dx = dp$

$$\Rightarrow I = \int \frac{(1+p^2)^2 dp}{1+p^6} = \int \frac{(1+p^2)}{p^4 - p^2 + 1} dp = \int \frac{p^2 \left(1 + \frac{1}{p^2}\right)}{p^2 \left(p^2 + \frac{1}{p^2} - 1\right)} dp$$

$$= \int \frac{dk}{k^2 + 1} = \tan^{-1}(k) + C \quad \left(\text{where } p - \frac{1}{p} = k, \left(1 + \frac{1}{p^2}\right) dp = dk \right)$$

$$= \tan^{-1}\left(p - \frac{1}{p}\right) + C = \tan^{-1}(\tan x - \cot x) + C = \tan^{-1}(-2\cot 2x) + C \quad \text{Ans. (C,D)}$$

Illustration 72: Evaluate: $\int \frac{2 \sin 2x - \cos x}{6 - \cos^2 x - 4 \sin x} dx$

Solution :
$$I = \int \frac{2 \sin 2x - \cos x}{6 - \cos^2 x - 4 \sin x} dx = \int \frac{(4 \sin x - 1) \cos x}{6 - (1 - \sin^2 x) - 4 \sin x} dx = \int \frac{(4 \sin x - 1) \cos x}{\sin^2 x - 4 \sin x + 5} dx$$

Put $\sin x = t$, so that $\cos x \, dx = dt$.

$$\therefore I = \int \frac{(4t-1)dt}{(t^2-4t+5)} \dots\dots (i)$$

Now, let $(4t - 1) = \lambda(2t - 4) + \mu$

Comparing coefficients of like powers of t , we get

$$\begin{aligned} 2\lambda &= 4, -4\lambda + \mu = -1 \\ \lambda &= 2, \mu = 7 \end{aligned} \quad \dots (ii)$$

$$\begin{aligned} \therefore I &= \int \frac{2(2t-4)+7}{t^2-4t+5} dt \quad \{\text{using (i) and (ii)}\} \\ &= 2 \int \frac{2t-4}{t^2-4t+5} dt + 7 \int \frac{dt}{t^2-4t+5} = 2 \log|t^2-4t+5| + 7 \int \frac{dt}{t^2-4t+4-4+5} \\ &= 2 \log|t^2-4t+5| + 7 \int \frac{dt}{(t-2)^2+(1)^2} = 2 \log|t^2-4t+5| + 7 \tan^{-1}(t-2) + C \\ &= 2 \log|\sin^2 x - 4 \sin x + 5| + 7 \tan^{-1}(\sin x - 2) + C. \end{aligned}$$

Ans.

Illustration 73: The value of $\int \sqrt{\frac{3-x}{3+x}} \cdot \sin^{-1}\left(\frac{1}{\sqrt{6}}\sqrt{3-x}\right) dx$, is equal to -

- (A) $\frac{1}{4} \left\{ -3 \left(\cos^{-1}\left(\frac{x}{3}\right) \right)^2 + 2\sqrt{9-x^2} \cdot \cos^{-1}\left(\frac{x}{3}\right) + 2x \right\} + C$
- (B) $\frac{1}{4} \left\{ -3 \left(\cos^{-1}\left(\frac{x}{3}\right) \right)^2 + 2\sqrt{9-x^2} \cdot \sin^{-1}\left(\frac{x}{3}\right) + 2x \right\} + C$
- (C) $\frac{1}{4} \left\{ -3 \left(\sin^{-1}\left(\frac{x}{3}\right) \right)^2 + 2\sqrt{9-x^2} \cdot \sin^{-1}\left(\frac{x}{3}\right) + 2x \right\} + C$
- (D) none of these

Solution :

$$\text{Here, } I = \int \sqrt{\frac{3-x}{3+x}} \cdot \sin^{-1}\left(\frac{1}{\sqrt{6}}\sqrt{3-x}\right) dx$$

$$\text{Put } x = 3\cos 2\theta \Rightarrow dx = -6\sin 2\theta d\theta$$

$$= \int \sqrt{\frac{3-3\cos 2\theta}{3+3\cos 2\theta}} \cdot \sin^{-1}\left(\frac{1}{\sqrt{6}}\sqrt{3-3\cos 2\theta}\right) (-6 \sin 2\theta) d\theta$$

$$= \int \frac{\sin \theta}{\cos \theta} \cdot \sin^{-1}(\sin \theta) \cdot (-6 \sin 2\theta) d\theta = -6 \int \theta \cdot (2 \sin^2 \theta) d\theta$$

$$= -6 \int \theta (1 - \cos 2\theta) d\theta = -6 \left\{ \frac{\theta^2}{2} - \int \theta \cos 2\theta d\theta \right\} + C$$

$$= -6 \left\{ \frac{\theta^2}{2} - \left(\theta \frac{\sin 2\theta}{2} - \int 1 \cdot \left(\frac{\sin 2\theta}{2} \right) d\theta \right) \right\} + C = -3\theta^2 + 6 \left\{ \theta \frac{\sin 2\theta}{2} + \frac{\cos 2\theta}{4} \right\} + C$$

$$= \frac{1}{4} \left\{ -3 \left(\cos^{-1}\left(\frac{x}{3}\right) \right)^2 + 2\sqrt{9-x^2} \cdot \cos^{-1}\left(\frac{x}{3}\right) + 2x \right\} + C$$

Ans. (A)

Illustration 74: Evaluate : $\int \frac{\tan\left(\frac{\pi}{4} - x\right)}{\cos^2 x \sqrt{\tan^3 x + \tan^2 x + \tan x}} dx$

Solution : $I = \int \frac{\tan\left(\frac{\pi}{4} - x\right)}{\cos^2 x \sqrt{\tan^3 x + \tan^2 x + \tan x}} dx = \int \frac{(1 - \tan^2 x) dx}{(1 + \tan x)^2 \cos^2 x \sqrt{\tan^3 x + \tan^2 x + \tan x}}$

$$I = \int \frac{-\left(1 - \frac{1}{\tan^2 x}\right) \sec^2 x dx}{\left(\tan x + 2 + \frac{1}{\tan x}\right) \sqrt{\tan x + 1 + \frac{1}{\tan x}}}$$

$$\text{let, } y = \sqrt{\tan x + 1 + \frac{1}{\tan x}} \Rightarrow 2y dy = \left(\sec^2 x - \frac{1}{\tan^2 x} \cdot \sec^2 x\right) dx$$

$$\therefore I = \int \frac{-2y dy}{(y^2 + 1) \cdot y} = -2 \int \frac{dy}{1 + y^2}$$

$$= -2 \tan^{-1} y + c = -2 \tan^{-1} \left(\sqrt{\tan x + 1 + \frac{1}{\tan x}} \right) + C$$

Ans.

Do yourself - 17 :

1. $\int \left[\left(\frac{x}{e}\right)^x + \left(\frac{e}{x}\right)^x \right] \ell n x dx$

2. Evaluate : $\int \frac{\tan 2\theta}{\sqrt{\cos^6 \theta + \sin^6 \theta}} d\theta$

3. Evaluate : $\int \frac{\sqrt{\cos \sec x - \cot x}}{\cos \sec x + \cot x} \cdot \frac{\sec x}{\sqrt{1 + 2 \sec x}} dx$

4. $\int \frac{dx}{\sqrt[4]{(x-1)^3 (x+2)^5}}$

5. $\int \frac{\cos x dx}{(1 - \cos x)^2}$

6. $\int \frac{(\cos 2x - 3) dx}{\cos^4 x \sqrt{4 - \cot^2 x}}$

7. $\int \frac{dx}{\sin \frac{x}{2} \sqrt{\cos^3 \frac{x}{2}}}$

8. $\int \frac{x^2 dx}{(1 + x^2)^2}$

9. $\int \frac{dx}{(1 + x^2)^4}$

10. $\int \frac{\sqrt{1 + x^2}}{x^4} dx$

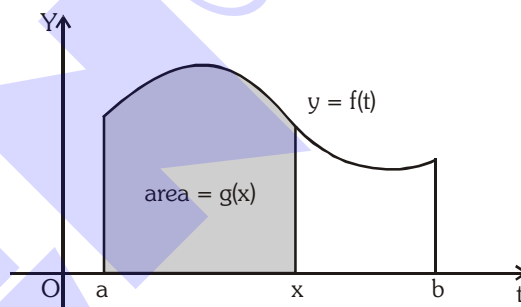
11. $\int \frac{x^2 - 1}{x^2 + 1} \cdot \frac{dx}{\sqrt{1 + x^4}}$

DEFINITE INTEGRATION

A definite integral is denoted by $\int_a^b f(x)dx$ which represent the algebraic area bounded by the curve $y = f(x)$, the ordinates $x = a$, $x = b$ and the x axis.

1. THE FUNDAMENTAL THEOREM OF CALCULUS :

The Fundamental Theorem of Calculus is appropriately named because it establishes a connection between the two branches of calculus : differential calculus and integral calculus. Differential calculus arose from the tangent problem, whereas integral calculus arose from a seemingly unrelated problem, the area problem. Newton's teacher at Cambridge, Isaac Barrow (1630-1677), discovered that these two problems are actually closely related. In fact, he realized that differentiation and integration are inverse processes. The Fundamental Theorem of Calculus gives the precise inverse relationship between the derivative and the integral. It was Newton and Leibnitz who exploited this relationship and used it to develop calculus into a systematic mathematical method. In particular, they saw that the Fundamental Theorem enabled them to compute areas and integrals very easily without having to compute them as limits of sums.



The Fundamental Theorem of Calculus, Part 1

If f is continuous on $[a, b]$, then the function g defined by

$$g(x) = \int_a^x f(t)dt \quad a \leq x \leq b$$

is continuous on $[a, b]$ and differentiable on (a, b) , and $g'(x) = f(x)$.

The Fundamental Theorem of Calculus, Part 2

If f is continuous on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

where F is any antiderivative of f , that is, a function such that $F' = f$.

Note : If $\int_a^b f(x)dx = 0 \Rightarrow$ then the equation $f(x) = 0$ has atleast one root lying in (a, b) provided f is a continuous function in (a, b) .

2. PROPERTIES OF DEFINITE INTEGRAL :

(a) $\int_a^b f(x)dx = \int_a^b f(t)dt$ provided f is same

$$(b) \int_a^b f(x) dx = - \int_b^a f(x) dx$$

(c) $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, where c may lie inside or outside the interval $[a, b]$. This property is to be used when f is piecewise continuous in (a, b) .

Illustration 1 : If $f(x) = \begin{cases} x^2, & 0 < x < 2 \\ 3x - 4, & 2 \leq x < 3 \end{cases}$ then evaluate $\int_0^3 f(x) dx$

Solution :

$$\begin{aligned} \int_0^3 f(x) dx &= \int_0^2 f(x) dx + \int_2^3 f(x) dx = \int_0^2 x^2 dx + \int_2^3 (3x - 4) dx \\ &= \left(\frac{x^3}{3} \right)_0^2 + \left(\frac{3x^2}{2} - 4x \right)_2^3 = \frac{8}{3} + \frac{27}{2} - 12 - 6 + 8 = 37/6 \end{aligned}$$

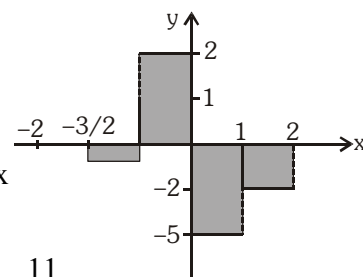
Ans.

Illustration 2 : If $f(x) = \begin{cases} 3[x] - 5 \frac{|x|}{x}, & x \neq 0 \\ 2, & x = 0 \end{cases}$ then $\int_{-3/2}^2 f(x) dx$ is equal to ($[.]$ denotes the greatest integer function)

(A) $-\frac{11}{2}$ (B) $-\frac{7}{2}$ (C) -6 (D) $-\frac{17}{2}$

Solution :

$$\begin{aligned} 3[x] - 5 \frac{|x|}{x} &= 3[x] - 5, \text{ if } x > 0 \\ &= 3[x] + 5, \text{ if } x < 0 \\ \Rightarrow \int_{-3/2}^2 f(x) dx &= \int_{-3/2}^{-1} (-1) dx + \int_{-1}^0 (2) dx + \int_0^1 (-5) dx + \int_1^2 (-2) dx \\ &= -1 \left(-1 + \frac{3}{2} \right) + 2(1) + 1(-5) + (-2) = -\frac{1}{2} + 2 - 5 - 2 = -\frac{11}{2} \end{aligned}$$



Ans. (A)

Illustration 3 : The value of $\int_1^2 (x^{[x^2]} + [x^2]^x) dx$, where $[.]$ denotes the greatest integer function, is equal to -

- (A) $\frac{5}{4} + \sqrt{3} + (2^{\sqrt{3}} - 2^{\sqrt{2}}) + \frac{1}{\log 3} (9 - 3^{\sqrt{3}})$
 (B) $\frac{5}{4} + \sqrt{3} + \frac{\sqrt{2}}{3} + \frac{1}{\log 2} (2^{\sqrt{3}} - 2^{\sqrt{2}}) + \frac{1}{\log 3} (9 - 3^{\sqrt{3}})$
 (C) $\frac{5}{4} + \frac{\sqrt{2}}{3} + \frac{1}{\log 2} (2^{\sqrt{3}} - 2^{\sqrt{2}}) + \frac{1}{\log 3} (9 - 3^{\sqrt{3}})$
 (D) none of these

7. Suppose that $F(x)$ is an antiderivative of $f(x) = \frac{\sin x}{x}$, $x > 0$ then $\int_1^3 \frac{\sin 2x}{x} dx$ can be expressed as-

- (A) $F(6) - F(2)$ (B) $\frac{1}{2}(F(6) - F(2))$ (C) $\frac{1}{2}(F(3) - F(1))$ (D) $2(F(6) - F(2))$

8. $\int_0^{\pi/2} |\sin x - \cos x| dx$

9. Evaluate: $\int_0^{\pi/2} \sqrt{\frac{1 - \sin 2x}{1 + \sin 2x}} dx$

Evaluate the following definite integrals.

10. $\int_0^{\ln 2} x e^{-x} dx$

11. $\int_{-1}^1 \frac{x dx}{\sqrt{5-4x}}$

12. $\int_0^{\pi/2} \frac{\cos x dx}{(1 + \sin x)(2 + \sin x)}$

13. $\int_0^{\pi/4} \cos 2x \sqrt{1 - \sin 2x} dx$

14. $\int_a^b \frac{|x|}{x} dx$

15. $\int_1^e \left(\frac{1}{\sqrt{x \ln x}} + \sqrt{\frac{\ln x}{x}} \right) dx$

16. $\int_2^e \left(\frac{1}{\ln x} - \frac{1}{\ln^2 x} \right) dx$

17. $\int_0^{\pi/4} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$

18. $\int_0^{\pi/4} \frac{\sin^2 x \cdot \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx$

19. $\int_0^{\pi/4} (x \cos x \cdot \cos 3x) dx$

20. $\int_0^{\pi/2} \frac{dx}{5 + 4 \sin x}$

21. $\int_2^3 \frac{dx}{(x-1)\sqrt{x^2-2x}}$

22. $\int_0^{\pi/2} \frac{dx}{1 + \cos \theta \cdot \cos x}$ $\theta \in (0, \pi)$

23. $\int_0^{\pi} \theta \sin^2 \theta \cos \theta d\theta$

24. $\int_0^{\pi/2} \frac{x + \sin x}{1 + \cos x} dx$

25. $\int_0^{3\pi/4} ((1+x)\sin x + (1-x)\cos x) dx$

26. $\int_0^{\pi} \left[\cos^2 \left(\frac{3\pi}{8} - \frac{x}{4} \right) - \cos^2 \left(\frac{11\pi}{8} + \frac{x}{4} \right) \right] dx$

27. $\int_0^{\pi/2} \sin \phi \cos \phi \sqrt{(a^2 \sin^2 \phi + b^2 \cos^2 \phi)} d\phi$ $a \neq b$ ($a > 0, b > 0$)

28. $\int_2^3 \frac{dx}{\sqrt{(x-1)(5-x)}}$

29. $\int_{3/2}^2 \left(\frac{x-1}{3-x} \right)^{1/2} dx$

30. $\int_0^{1/2} \frac{dx}{(1-2x^2)\sqrt{1-x^2}}$

31. $\int_1^2 \frac{dx}{x(x^4+1)}$

32. $\int_0^1 \frac{dx}{x^2 + 2x \cos \alpha + 1}$ where $-\pi < \alpha < \pi$

33. $\int_a^b \frac{dx}{\sqrt{1+x^2}}$ where $a = \frac{e-e^{-1}}{2}$ & $b = \frac{e^2-e^{-2}}{2}$

34. $\int_0^1 \frac{1-x^2}{1+x^2+x^4} dx$

35. $\int_0^{\pi/4} \frac{\sin \theta + \cos \theta}{9 + 16 \sin 2\theta} d\theta$

36. $\int_{-1}^1 \left(\frac{d}{dx} \left(\frac{1}{1 + e^{1/x}} \right) \right) dx$

37. $\int_1^e \frac{dx}{\ln(x^x e^x)}$

38. If $f(\pi) = 2$ & $\int_0^{\pi} (f(x) + f''(x)) \sin x \, dx = 5$, then find $f(0)$

39. $\int_{\ell n 2}^{\ell n 3} f(x) dx$, where $f(x) = e^{-x} + 2e^{-2x} + 3e^{-3x} + \dots \infty$ 40.

$$\int_0^{\pi/2} \sqrt{\frac{\sec x - \tan x}{\sec x + \tan x}} \cdot \frac{\operatorname{cosec} x}{\sqrt{1 + 2 \operatorname{cosec} x}} dx$$

41. $\int_0^1 x f''(x) \, dx$, where $f(x) = \cos(\tan^{-1} x)$

42. $\int_{\pi/2}^{\pi} x^{\sin x} (1 + x \cos x \cdot \ln x + \sin x) dx$

43. Given $f'(x) = \frac{\cos x}{x}$, $f\left(\frac{\pi}{2}\right) = a$, $f\left(\frac{3\pi}{2}\right) = b$. Find the value of the definite integral $\int_{\pi/2}^{3\pi/2} f(x) dx$.

44. Suppose that f , f' and f'' are continuous on $[0, \ln 2]$ and that $f(0) = 0$, $f'(0) = 3$, $f(\ln 2) = 6$, $f'(\ln 2) = 4$ and $\int_0^{\ln 2} e^{-2x} \cdot f(x) dx = 3$. Find the value of $\int_0^{\ln 2} e^{-2x} \cdot f''(x) dx$.

45. Suppose that the function f , g , f' and g' are continuous over $[0, 1]$, $g(x) \neq 0$ for $x \in [0, 1]$,
 $f(0) = 0$, $g(0) = \pi$, $f(1) = \frac{2009}{2}$ and $g(1) = 1$.

Find the value of the definite integral, $\int_0^1 \frac{f(x) \cdot g'(x) \{g^2(x) - 1\} + f'(x) \cdot g(x) \{g^2(x) + 1\}}{g^2(x)} dx$.

(d) $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx = \begin{cases} 0 & \text{; if } f(x) \text{ is an odd function} \\ 2 \int_0^a f(x) dx & \text{; if } f(x) \text{ is an even function} \end{cases}$

Illustration 5 : Evaluate $\int_{-1/2}^{1/2} \cos x \ln\left(\frac{1+x}{1-x}\right) dx$

Solution : $f(-x) = \cos(-x) \ln\left(\frac{1-x}{1+x}\right) = -\cos \ln\left(\frac{1+x}{1-x}\right) = -f(x)$

$$\Rightarrow f(x) \text{ is odd}$$

Hence, the value of the given integral = 0.

Ans.

Illustration 6 : If $f(x) = \begin{vmatrix} \cos x & e^{x^2} & 2x \cos^2 x / 2 \\ x^2 & \sec x & \sin x + x^3 \\ 1 & 2 & x + \tan x \end{vmatrix}$, then the value of $\int_{-\pi/2}^{\pi/2} (x^2 + 1)(f(x) + f''(x))dx$

(A) 1

(B) -1

(C) 2

(D) none of these

Solution : As, $f(x) = \begin{vmatrix} \cos x & e^{x^2} & 2x \cos^2 x / 2 \\ x^2 & \sec x & \sin x + x^3 \\ 1 & 2 & x + \tan x \end{vmatrix}$

$$\Rightarrow f(-x) = -f(x) \quad \Rightarrow f(x) \text{ is odd}$$

$$\Rightarrow f(x) \text{ is even} \quad \Rightarrow f''(x) \text{ is odd}$$

Thus, $f(x) + f''(x)$ is odd function let,

$$\phi(x) = (x^2 + 1) \cdot \{f(x) + f''(x)\}$$

$$\Rightarrow \phi(-x) = -\phi(x)$$

i.e. $\phi(x)$ is odd

$$\therefore \int_{-\pi/2}^{\pi/2} \phi(x)dx = 0$$

Ans. (D)

Do yourself - 2 :

Evaluate :

1. $\int_{-\pi/2}^{\pi/2} (x^2 \sin^3 x + \cos x)dx$

2. $\int_{-\pi/2}^{\pi/2} \ln \left[2 \left(\frac{4 - \sin \theta}{4 + \sin \theta} \right) \right] d\theta$

3. Evaluate : $\int_{-2}^2 \frac{x^2 - x}{\sqrt{x^2 + 4}} dx$

4. Evaluate : $\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7 + 3x^6 - 10x^5 - 7x^3 - 12x^2 + x + 1}{x^2 + 2} dx$

5. Evaluate : $\int_{-1}^1 \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1 + x^{666}} dx$

6. The value of the integral $\int_{-1}^1 \log(x + \sqrt{x^2 + 1}) dx$ is :

(A) 2

(B) 0

(C) -1

(D) 1

(e) $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$, In particular $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

Illustration 7 : If f, g, h be continuous functions on $[0, a]$ such that $f(a-x) = -f(x)$, $g(a-x) = g(x)$ and

$3h(x) - 4h(a-x) = 5$, then prove that $\int_0^a f(x)g(x)h(x)dx = 0$

Solution : $I = \int_0^a f(x)g(x)h(x)dx = \int_0^a f(a-x)g(a-x)h(a-x)dx = -\int_0^a f(x)g(x)h(a-x)dx$

$7I = 3I + 4I$

$= \int_0^a f(x)g(x)\{3h(x) - 4h(a-x)\}dx = 5 \int_0^a f(x)g(x)dx = 0$

(since $f(a-x)g(a-x) = -f(x)g(x)$)

$\Rightarrow I = 0$

Ans.

Illustration 8 : Evaluate $\int_{-\pi}^{\pi} \frac{x \sin x}{e^x + 1} dx$

Solution : $I = \int_{-\pi}^0 \frac{x \sin x}{e^x + 1} dx + \int_0^{\pi} \frac{x \sin x}{e^x + 1} dx = I_1 + I_2$

where $I_1 = \int_{-\pi}^0 \frac{x \sin x}{e^x + 1} dx$

Put $x = -t \Rightarrow dx = -dt$

$\Rightarrow I_1 = \int_{\pi}^0 \frac{(-t) \sin(-t)(-dt)}{e^{-t} + 1} = \int_0^{\pi} \frac{t \sin t dt}{e^{-t} + 1} = \int_0^{\pi} \frac{e^t t \sin t dt}{e^t + 1} = \int_0^{\pi} \frac{e^x x \sin x dx}{e^x + 1}$

Hence $I = I_1 + I_2 = \int_0^{\pi} \frac{e^x x \sin x}{e^x + 1} dx + \int_0^{\pi} \frac{x \sin x}{e^x + 1} dx$

$I = \int_0^{\pi} x \sin x dx = \int_0^{\pi} (\pi - x) \sin(\pi - x) dx = \pi \int_0^{\pi} \sin x dx - I$

$\Rightarrow 2I = \pi \int_0^{\pi} \sin x dx = \pi [-\cos x]_0^{\pi} = 2\pi \Rightarrow I = \pi$

Ans.

Illustration 9 : Evaluate $\int_0^2 \frac{dx}{(17+8x-4x^2)[e^{6(1-x)}+1]}$

Solution : Let $I = \int_0^2 \frac{dx}{(17+8x-4x^2)[e^{6(1-x)}+1]}$

$$\text{Also } I = \int_0^2 \frac{dx}{(17+8x-4x^2)[e^{-6(1-x)}+1]} \left[\because \int_0^a f(x)dx = \int_0^a f(a-x)dx \right]$$

Adding, we get

$$\begin{aligned} 2I &= \int_0^2 \frac{1}{17+8x-4x^2} \left(\frac{1}{e^{6(1-x)}+1} + \frac{1}{e^{-6(1-x)}+1} \right) dx \\ &= \int_0^2 \frac{1}{17+8x-4x^2} dx = -\frac{1}{4} \int_0^2 \frac{dx}{x^2-2x-17/4} \\ &= -\frac{1}{4} \int_0^2 \frac{dx}{(x-1)^2-21/4} = -\frac{1}{4} \times \frac{1}{2 \times \frac{\sqrt{21}}{2}} \left[\log \left| \frac{x-1-\frac{\sqrt{21}}{2}}{x-1+\frac{\sqrt{21}}{2}} \right| \right]_0^2 \\ &= -\frac{1}{4\sqrt{21}} \left[\log \left| \frac{2x-2-\sqrt{21}}{2x-2+\sqrt{21}} \right| \right]_0^2 \Rightarrow I = -\frac{1}{8\sqrt{21}} \left[\log \left| \frac{2-\sqrt{21}}{2+\sqrt{21}} \right| - \log \left| \frac{2+\sqrt{21}}{\sqrt{21}-2} \right| \right] \\ &= -\frac{1}{4\sqrt{21}} \left[\log \left| \frac{\sqrt{21}-2}{2+\sqrt{21}} \right| \right] \end{aligned}$$

Ans.

Illustration 10 : $\int_0^1 \cot^{-1}(1-x+x^2)dx$ equals -

- (A) $\frac{\pi}{2} + \log 2$ (B) $\frac{\pi}{2} - \log 2$ (C) $\pi - \log 2$ (D) none of these

Solution : $I = \int_0^1 \tan^{-1} \left(\frac{1}{1-x+x^2} \right) dx = \int_0^1 \tan^{-1} \left(\frac{x+(1-x)}{1-x(1-x)} \right) dx$

$$= \int_0^1 [\tan^{-1} x + \tan^{-1}(1-x)] dx = \int_0^1 \tan^{-1} x dx + \int_0^1 \tan^{-1}(1-x) dx$$

$$= 2 \int_0^1 \tan^{-1} x dx = 2 \left[x \tan^{-1} x - \frac{1}{2} \log(1+x^2) \right]_0^1 = 2 \frac{\pi}{4} - \log 2 = \frac{\pi}{2} - \log 2 \quad \text{Ans. (B)}$$

Illustration 11 : $\int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx$

Solution :
$$I = \int_0^{\pi/2} \frac{a \sin x + b \cos x}{\sin x + \cos x} dx \quad \dots (i)$$

$$I = \int_0^{\pi/2} \frac{a \sin(\pi/2 - x) + b \cos(\pi/2 - x)}{\sin(\pi/2 - x) + \cos(\pi/2 - x)} dx = \int_0^{\pi/2} \frac{a \cos x + b \sin x}{\sin x + \cos x} dx \quad \dots (ii)$$

$$\therefore 2I = \int_0^{\pi/2} \frac{(a+b)(\sin x + \cos x)}{\sin x + \cos x} dx = \int_0^{\pi/2} (a+b) dx = (a+b)\pi/2 \Rightarrow I = (a+b)\pi/4 \quad \text{Ans.}$$

Illustration 12: $\int_0^{\pi/2} \frac{2^{\sin x}}{2^{\sin x} + 2^{\cos x}} dx$ equals -

(A) 2 (B) π (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Solution :
$$I = \int_0^{\pi/2} \frac{2^{\sin x}}{2^{\sin x} + 2^{\cos x}} dx = \int_0^{\pi/2} \frac{2^{\sin(\pi/2-x)}}{2^{\sin x(\pi/2-x)} + 2^{\cos(\pi/2-x)}} dx = \int_0^{\pi/2} \frac{2^{\cos x}}{2^{\cos x} + 2^{\sin x}} dx$$

$$2I = \int_0^{\pi/2} dx = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4} \quad \text{Ans. (C)}$$

Do yourself - 3 :

Evaluate :

1. $\int_1^5 \frac{\sqrt{x}}{\sqrt{6-x} + \sqrt{x}} dx$

2. $\int_{\pi/6}^{\pi/3} \frac{dx}{1 + \tan^5 x}$

3. $\int_0^3 \sqrt{\frac{x}{3-x}} \, dx$

4. $\int_1^3 \frac{x^2}{x^2 - 4x + 8} dx$ is equal to -

(A) 2 (B) 1

(C) $2 \tan^{-1} \frac{1}{2}$

(D) $2\left(1 + \tan^{-1} \frac{1}{2}\right)$

5. The integral $\int_0^1 x(1-x)^9 dx$ equals to

(A) $\frac{1}{10}$ (B) $\frac{1}{90}$

(C) $\frac{1}{72}$

(D) $\frac{1}{110}$

6. Prove that :

$$\textbf{(a)} \quad \int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}} dx = (\beta-\alpha) \frac{\pi}{2}$$

(b) $\int_{\alpha}^{\beta} \frac{x \cdot dx}{\sqrt{(x-\alpha)(\beta-x)}} = (\alpha + \beta) \frac{\pi}{2}$, where $\alpha < \beta$

7. Evaluate: $\int_{1/3}^3 \frac{\sin^{-1} \frac{x}{\sqrt{1+x^2}}}{x} dx$

8. The value of the definite integral $\int_{\pi/24}^{5\pi/24} \frac{dx}{1 + \sqrt[3]{\tan 2x}}$ is :
- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{18}$
9. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1 + \sin^2 x}{1 + \pi^{\sin x}} \right) dx$ is
- (A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$
10. $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to :
- (A) 6 (B) 8 (C) 5 (D) 10
11. The value of $\int_{-\pi/2}^{\pi/2} \frac{\cos^2 x}{1 + 3^x} dx$ is
- (A) $\frac{\pi}{4}$ (B) 4π (C) $\frac{\pi}{2}$ (D) 2π
12. $\int_0^{\pi/2n} \frac{dx}{1 + \tan^n nx} =$
- (A) 0 (B) $\frac{\pi}{4n}$ (C) $\frac{n\pi}{4}$ (D) $\frac{\pi}{2n}$
13. Evaluate : $\int_0^{\pi} \frac{(ax + b) \sec x \tan x}{4 + \tan^2 x} dx (a, b > 0)$
14. Evaluate : $\int_0^{\pi} \frac{(2x + 3) \sin x}{(1 + \cos^2 x)} dx$
15. $\int_{-\pi/3}^{-\pi/6} (\sin^6 \theta + \frac{3}{2} \sin^2 \theta \cos^2 \theta) d\theta$ equals :-
- (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{2}$ (C) 1 (D) $\frac{1}{2}$
16. If $\int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x dx}{(\pi^2 - 3\pi x + 3x^2)} = k \int_0^{\pi/2} \sin^2 x dx$, then the value of k is :-
- (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{16}$ (D) $\frac{\pi}{8}$

$$(f) \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx = \begin{cases} 2 \int_0^a f(x) dx & ; \text{ if } f(2a - x) = f(x) \\ 0 & ; \text{ if } f(2a - x) = -f(x) \end{cases}$$

Illustration 13: Evaluate $\int_0^{\pi} \frac{x dx}{1 + \cos^2 x}$

Solution : Let $I = \int_0^{\pi} \frac{x dx}{1 + \cos^2 x} = \int_0^{\pi} \frac{(\pi - x) dx}{1 + \cos^2(\pi - x)} = \int_0^{\pi} \frac{\pi dx}{1 + \cos^2 x} - I$

$$\Rightarrow 2I = \int_0^{\pi} \frac{\pi dx}{1 + \cos^2 x} = 2\pi \int_0^{\pi/2} \frac{dx}{1 + \cos^2 x} = 2\pi \int_0^{\pi/2} \frac{\sec^2 x dx}{2 + \tan^2 x}$$

Let $\tan x = t$ so that for $x \rightarrow 0$, $t \rightarrow 0$ and for $x \rightarrow \pi/2$, $t \rightarrow \infty$. Hence we can write,

$$I = \pi \int_0^{\infty} \frac{dt}{2 + t^2} = \pi \frac{1}{\sqrt{2}} \left[\tan^{-1} \frac{t}{\sqrt{2}} \right]_0^{\infty} = \frac{\pi^2}{2\sqrt{2}}$$

Ans.

Illustration 14: Prove that $\int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\cos x) dx = -\frac{\pi}{2} \log 2$

Solution : Let $I = \int_0^{\pi/2} \log(\sin x) dx$ (i)

$$\text{then } I = \int_0^{\pi/2} \log \sin \left(\frac{\pi}{2} - x \right) dx = \int_0^{\pi/2} \log(\cos x) dx \quad \text{..... (ii)}$$

adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \log \sin x dx + \int_0^{\pi/2} \log \cos x dx = \int_0^{\pi/2} (\log \sin x + \log \cos x) dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \log(\sin x \cos x) dx = \int_0^{\pi/2} \log \left(\frac{2 \sin x \cos x}{2} \right) dx$$

$$= \int_0^{\pi/2} \log \left(\frac{\sin 2x}{2} \right) dx = \int_0^{\pi/2} \log(\sin 2x) dx - \int_0^{\pi/2} (\log 2) dx = \int_0^{\pi/2} \log \sin 2x \cdot dx - (\log 2)(x)_0^{\pi/2}$$

$$\Rightarrow 2I = \int_0^{\pi/2} \log(\sin 2x) dx - \frac{\pi}{2} \log 2 \quad \text{..... (iii)}$$

Let $I_1 = \int_0^{\pi/2} \log(\sin 2x) dx$, putting $2x = t$, we get

$$I_1 = \int_0^{\pi} \log(\sin t) \frac{dt}{2} = \frac{1}{2} \int_0^{\pi} \log(\sin t) dt = \frac{1}{2} \cdot 2 \int_0^{\pi/2} \log(\sin t) dt$$

$$I_1 = \int_0^{\pi/2} \log(\sin x) dx$$

$$\therefore \text{ (iii) becomes ; } 2I = I - \frac{\pi}{2} \log 2$$

$$\text{Hence } \int_0^{\pi/2} \log \sin x dx = -\frac{\pi}{2} \log 2$$

Ans.

Illustration 15 : $\int_0^{\pi/2} (2 \log \sin x - \log \sin 2x) dx$ equals -

- (A) $\pi \log 2$ (B) $-\pi \log 2$ (C) $(\pi/2) \log 2$ (D) $-(\pi/2) \log 2$

Solution :
$$I = \int_0^{\pi/2} (2 \log \sin x - \log 2 \sin x \cos x) dx = \int_0^{\pi/2} (2 \log \sin x - \log 2 - \log \sin x - \log \cos x) dx$$

$$= \int_0^{\pi/2} \log \sin x dx - \int_0^{\pi/2} \log 2 dx - \int_0^{\pi/2} \log \cos x dx = -(\pi/2) \log 2$$
 Ans. (D)

Do yourself - 4 :

Evaluate :

1. $\int_{-\sqrt{3}}^{\sqrt{3}} \frac{dx}{(1+e^x)(1+x^2)}$
2. $\int_0^{\pi/2} \ln(\sin^2 x \cos x) dx$
3. $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx$
4. $\int_{-\pi/2}^{\pi/2} \sqrt{\cos x - \cos^3 x} dx$
5. The value of $\int_{-1/\sqrt{2}}^{1/\sqrt{2}} \left(\left(\frac{x+1}{x-1} \right)^2 + \left(\frac{x-1}{x+1} \right)^2 - 2 \right)^{1/2} dx$ is :
 (A) $\log_e 4$ (B) $\log_e 16$ (C) $2 \log_e 16$ (D) $4 \log_e (3 + 2\sqrt{2})$

(g) $\int_0^{nT} f(x) dx = n \int_0^T f(x) dx$, ($n \in I$); where 'T' is the period of the function i.e. $f(T+x) = f(x)$

Note that : $\int_x^{T+x} f(t) dt$ will be independent of x and equal to $\int_0^T f(t) dt$

(h) $\int_{a+nT}^{b+nT} f(x) dx = \int_a^b f(x) dx$ where $f(x)$ is periodic with period T & $n \in I$.

(i) $\int_{mT}^{nT} f(x) dx = (n-m) \int_0^T f(x) dx$, ($n, m \in I$) if $f(x)$ is periodic with period 'T'.

Illustration 16: Evaluate $\int_0^{4\pi} |\cos x| dx$

Solution : Note that $|\cos x|$ is a periodic function with period π . Hence the given integral.

$$I = 4 \int_0^{\pi} |\cos x| dx = 4 \left[\int_0^{\frac{\pi}{2}} \cos x dx - \int_{\frac{\pi}{2}}^{\pi} \cos x dx \right] = 4 \left[[\sin x]_0^{\pi/2} - [\sin x]_{\pi/2}^{\pi} \right] = 4[1+1] = 8 \quad \text{Ans.}$$

Illustration 17: Evaluate $\int_0^{16\pi/3} |\sin x| dx$

Solution :

$$\begin{aligned} \int_0^{16\pi/3} |\sin x| dx &= \int_0^{5\pi} |\sin x| dx + \int_{5\pi}^{5\pi+\pi/3} |\sin x| dx = 5 \int_0^{\pi} |\sin x| dx + \int_0^{\pi/3} |\sin x| dx \\ &= 5[-\cos x]_0^{\pi} + [-\cos x]_0^{\pi/3} = 10 + \left(-\frac{1}{2} + 1\right) = \frac{21}{2} \quad \text{Ans.} \end{aligned}$$

Illustration 18: Evaluate : $\int_0^{2n\pi} [\sin x + \cos x] dx$. Here $[.]$ is the greatest integer function.

Solution : Let $I = \int_0^{2n\pi} [\sin x + \cos x] dx = n \int_0^{2\pi} [\sin x + \cos x] dx$
 $(\because [\sin x + \cos x] \text{ is periodic function with period } 2\pi)$

$$[\sin x + \cos x] = \begin{cases} 1, & 0 \leq x \leq \frac{\pi}{2} \\ 0, & \frac{\pi}{2} \leq x \leq \frac{3\pi}{4} \\ -1, & \frac{3\pi}{4} < x \leq \pi \\ -2, & \pi < x \leq \frac{3\pi}{2} \\ -1, & \frac{3\pi}{2} < x \leq \frac{7\pi}{4} \\ 0, & \frac{7\pi}{4} < x \leq 2\pi \end{cases}$$

$$\text{Hence } I = n \left[\int_0^{\pi/2} 1 dx + \int_{\pi/2}^{3\pi/4} 0 dx + \int_{3\pi/4}^{\pi} -1 dx + \int_{\pi}^{3\pi/2} -2 dx + \int_{3\pi/2}^{7\pi/4} -1 dx + \int_{7\pi/4}^{2\pi} 0 dx \right]$$

$$I = n \left[\frac{\pi}{2} + 0 - \pi + \frac{3\pi}{4} - 3\pi + 2\pi - \frac{7\pi}{4} + \frac{3\pi}{2} + 0 \right] = -n\pi \quad \text{Ans.}$$

Do yourself - 5 :**Evaluate :**

- $\int_{-1.5}^{10} \{2x\} dx$, where $\{.\}$ denotes fractional part of x .
- $\int_{20\pi + \frac{\pi}{6}}^{20\pi + \frac{\pi}{3}} (\sin x + \cos x) dx$
- If $g(x) = \int_0^x \cos^4 t dt$, then $g(x + \pi)$ equals
(A) $g(x) + g(\pi)$ (B) $g(x) - g(\pi)$ (C) $g(x) g(\pi)$ (D) $[g(x)/g(\pi)]$
- The value of $\int_{-1}^1 x^2 e^{[x^3]} dx$, where $[t]$ denotes the greatest integer $\leq t$, is :
(A) $\frac{e-1}{3e}$ (B) $\frac{e+1}{3}$ (C) $\frac{e+1}{3e}$ (D) $\frac{1}{3e}$
- The value of $\sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx$, where $[x]$ is the greatest integer $\leq x$, is
(A) $100(e-1)$ (B) $100(1-e)$ (C) $100e$ (D) $100(1+e)$
- If the value of the integral $\int_0^5 \frac{x+[x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta$, where $\alpha, \beta \in \mathbb{R}$, $5\alpha + 6\beta = 0$, and $[x]$ denotes the greatest integer less than or equal to x ; then the value of $(\alpha + \beta)^2$ is equal to :
(A) 100 (B) 25 (C) 16 (D) 36
- $\int_{-3}^3 \{\sin\{x\}\} dx$ is equal to (where $\{.\}$ represent fractional part of x)
(A) $1 - \cos 1$ (B) $3(1 - \cos 1)$ (C) $6(1 - \cos 1)$ (D) 0
- The value of $I = \left| \int_{-3}^3 \ln \left(\frac{(\sqrt{1+x+x^2}(x-[x]))}{(\sqrt{1-x+x^2})} \right) dx \right|$ is (where $[.]$ denotes greatest integer function)
- If $f(x)$ is an even function which is also periodic with the period T and $\int_0^a f(x) dx = 3$ and $\int_{-T/2}^{3T/2} f(x) dx = 18$, then $\int_{-a}^{a+5T} f(x) dx$ is equal to -
(A) 96 (B) 93 (C) 51 (D) 48
- Let $f(x) = \begin{cases} -2x-8 & -4 < x \leq -2 \\ -x^2 & -2 < x \leq 0 \\ x^2 & 0 < x \leq 2 \\ 8-2x & 2 < x \leq 4 \end{cases}$ and $f(x+8) = f(x)$ for all $x \in \mathbb{R}$. Then which of the following results are true ?
 $R_1 = \int_0^4 f(x) dx = \frac{20}{3}$; $R_2 = \int_2^{10} f(x) dx = 0$; $R_3 = \int_{-4}^4 f(x) dx = \frac{40}{3}$; $R_4 = \int_0^8 f(x) dx = 0$
(A) only R_1 is true (B) only R_3 is true (C) only R_2 & R_4 are true (D) R_1, R_2, R_4 are true

3. WALLI'S FORMULA :

If $m, n \in \mathbb{N}$ & $m, n \geq 2$, then

$$(a) \int_0^{\pi/2} \sin^n x \, dx = \int_0^{\pi/2} \cos^n x \, dx = \frac{(n-1)(n-3)\dots(1 \text{ or } 2)}{n(n-2)\dots(1 \text{ or } 2)} K$$

$$\text{where } K = \begin{cases} \pi/2 & \text{if } n \text{ is even} \\ 1 & \text{if } n \text{ is odd} \end{cases}$$

$$(b) \int \sin^n x \cdot \cos^m x \, dx = \frac{[(n-1)(n-3)(n-5)\dots 1 \text{ or } 2][(m-1)(m-3)\dots 1 \text{ or } 2]}{(m+n)(m+n-2)(m+n-4)\dots 1 \text{ or } 2} K$$

$$\text{Where } K = \begin{cases} \frac{\pi}{2} & \text{if both } m \text{ and } n \text{ are even} \\ 1 & \text{otherwise} \end{cases}$$

Illustration 19: $\int_{-\pi/2}^{\pi/2} \sin^4 x \cos^6 x \, dx =$

(A) $\frac{3\pi}{64}$ (B) $\frac{3\pi}{572}$ (C) $\frac{3\pi}{256}$ (D) $\frac{3\pi}{128}$

Solution : $I = \int_{-\pi/2}^{\pi/2} \sin^4 x \cos^6 x \, dx = 2 \int_0^{\pi/2} \sin^4 x \cos^6 x \, dx = 2 \frac{(3.1)(5.3.1)}{10.8.6.4.2} \cdot \frac{\pi}{2} = \frac{3\pi}{256}$ **Ans. (C)**

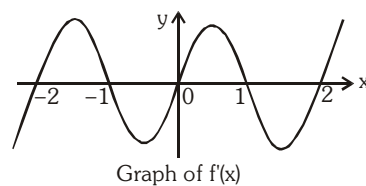
4. DERIVATIVE OF ANTIDERIVATIVE FUNCTION (Newton-Leibnitz Formula) :

If $h(x)$ & $g(x)$ are differentiable functions of x then, $\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt = f[h(x)] \cdot h'(x) - f[g(x)] \cdot g'(x)$

Illustration 20: Find the points of maxima/minima of $\int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$

Solution : Let $f(x) = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t} dt$

$$f'(x) = \frac{x^4 - 5x^2 + 4}{2 + e^{x^2}} \cdot 2x = 0 = \frac{(x-1)(x+1)(x-2)(x+2)2x}{2 + e^{x^2}}$$



From the wavy curve, it is clear that $f(x)$ changes its sign at $x = \pm 2, \pm 1, 0$ and hence the points of maxima are $-1, 1$ and of the minima are $-2, 0, 2$.

Illustration 21 : Evaluate $\frac{d}{dt} \int_{t^2}^{t^3} \frac{1}{\log x} dx$

Solution : $\frac{d}{dt} \int_{t^2}^{t^3} \frac{1}{\log x} dx = \frac{1}{\log t^3} \cdot \frac{d}{dt}(t^3) - \frac{1}{\log t^2} \cdot \frac{d}{dt}(t^2) = \frac{3t^2}{3 \log t} - \frac{2t}{2 \log t} = \frac{t(t-1)}{\log t}$

Ans.

Do yourself - 6 :

1. If $\int_0^{\pi/4} \sin^6 x \cos^4 x dx = \frac{3\pi}{1024} - \frac{K}{960}$, then K is equal to

2. If $f(x) = \int_{1/x}^{\sqrt{x}} \sin t dt$, then find $f'(1)$.

3. $\int_{\pi/3}^x \sqrt{3 - \sin^2 t} dt + \int_0^y \cos t dt = 0$, then evaluate $\frac{dy}{dx}$.

4. If $g(x) = \int_1^x e^{t^2} dt$ then the value of $\int_3^{x^3} e^{t^2} dt$ equals

- (A) $g(x^3) - g(3)$ (B) $g(x^3) + g(3)$ (C) $g(x^3) - 3$ (D) $g(x^3) - 3g(x)$

5. $\lim_{x \rightarrow 0} \frac{\int_0^x \sin t^2 dt}{x(1 - \cos x)}$ equals -

- (A) $\frac{1}{3}$ (B) 2 (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

6. $\lim_{h \rightarrow 0} \frac{\int_a^{x+h} \ln^2 t dt - \int_a^x \ln^2 t dt}{h} =$

- (A) 0 (B) $\ln^2 x$ (C) $\frac{2 \ln x}{x}$ (D) does not exist

7. $\lim_{k \rightarrow 0} \frac{1}{k} \int_0^k (1 + \sin 2x)^{\frac{1}{x}} dx =$

- (A) 2 (B) 1 (C) e^2 (D) non existent

8. Find the constants 'a' ($a > 0$) and 'b' such that $\lim_{x \rightarrow 0} \frac{\int_0^x \frac{t^2 dt}{\sqrt{a+t}}}{bx - \sin x} = 1$.

9. Let $f : (a, b) \rightarrow \mathbb{R}$ be twice differentiable function such that $f(x) = \int_a^x g(t) dt$ for a differentiable function $g(x)$. If $f(x) = 0$ has exactly five distinct roots in (a, b) , then $g(x)g'(x) = 0$ has at least :
- (A) twelve roots in (a, b) (B) five roots in (a, b)
(C) seven roots in (a, b) (D) three roots in (a, b)
10. Let $y = f(x)$ be a differentiable curve satisfying $\int_2^x f(t) dt = \frac{x^2}{2} + \int_x^2 t^2 f(t) dt$, then
- $\int_{-\pi/4}^{\pi/4} \frac{f(x) + x^9 - x^3 + x + 1}{\cos^2 x} dx$ equals -
- (A) 0 (B) 1 (C) 2 (D) 4
11. If $\int_0^{g(x)} f(t) dt = x^2 + \cos \pi x + 1 \quad \forall \quad x \geq 1$, where $g(x)$ is inverse of $f(x)$. If $f(3) = 4$, then $f'(3)$ is
- (A) $\frac{1}{2}$ (B) $\frac{9}{4}$ (C) 8 (D) 4
12. $\int_0^{\pi/4} (\cos 2x)^{3/2} \cdot \cos x dx =$
- (A) $\frac{3\pi}{16}$ (B) $\frac{3\pi}{32}$ (C) $\frac{3\pi}{16\sqrt{2}}$ (D) $\frac{3\pi\sqrt{2}}{16}$

5. DEFINITE INTEGRAL AS LIMIT OF A SUM :

An alternative way of describing $\int_a^b f(x)dx$ is that the definite integral $\int_a^b f(x)dx$ is a limiting case of the summation of an infinite series, provided $f(x)$ is continuous on $[a,b]$

i.e. $\int_a^b f(x)dx = \lim_{n \rightarrow \infty} h \sum_{r=0}^{n-1} f(a+rh)$ where $h = \frac{b-a}{n}$. The converse is also true i.e., if we have an infinite series of the above form, it can be expressed as a definite integral.

Step I: Express the given series in the form $\sum \frac{1}{n} f\left(\frac{r}{n}\right)$

Step II : Then the limit is its sum when $n \rightarrow \infty$, i.e. $\lim_{n \rightarrow \infty} \frac{1}{n} f\left(\frac{r}{n}\right)$

Step III : Replace $\frac{r}{n}$ by x and $\frac{1}{n}$ by dx and $\lim_{n \rightarrow \infty} \sum$ by the sign of $\int$

Step IV : The lower and the upper limit of integration are the limiting values of $\frac{r}{n}$ for the first and the last term of r respectively.

Illustration 22 : Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1}{2n+1} + \frac{1}{2n+2} + \dots + \frac{1}{6n} \right)$

Solution : Let $S_n = \frac{1}{2n+1} + \frac{1}{2n+2} + \dots + \frac{1}{6n} = \sum_{r=1}^{4n} \frac{1}{2n+r} = \sum_{r=1}^{4n} \frac{1}{n} \cdot \frac{1}{2 + \left(\frac{r}{n}\right)}$

$$\Rightarrow S = \lim_{n \rightarrow \infty} S_n = \int_0^4 \frac{dx}{2+x} = [\ln |2+x|]_0^4 = \ln 6 - \ln 2 = \ln 3$$

Ans.

Illustration 23 : Evaluate $\lim_{n \rightarrow \infty} \left[\frac{\sqrt{n}}{(3+4\sqrt{n})^2} + \frac{\sqrt{n}}{\sqrt{2}(3\sqrt{2}+4\sqrt{n})^2} + \frac{\sqrt{n}}{\sqrt{3}(3\sqrt{3}+4\sqrt{n})^2} + \dots + \frac{1}{49n} \right]$

Solution : Let $p = \lim_{n \rightarrow \infty} \left[\frac{\sqrt{n}}{(3+4\sqrt{n})^2} + \frac{\sqrt{n}}{\sqrt{2}(3\sqrt{2}+4\sqrt{n})^2} + \dots + \frac{\sqrt{n}}{\sqrt{n}(3\sqrt{n}+4\sqrt{n})^2} \right]$

Analyzing the expression with the view of increasing integral value we get the expression in terms of r as

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r}+4\sqrt{n})^2} = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n\sqrt{\frac{r}{n}} \left(3\sqrt{\frac{r}{n}} + 4 \right)^2} = \int_0^1 \frac{dx}{\sqrt{x}(3\sqrt{x}+4)^2}$$

Put $3\sqrt{x} + 4 = t$, $\therefore \frac{3}{2\sqrt{x}} dx = dt$

Hence $p = \frac{2}{3} \int_4^7 \frac{dt}{t^2} = \frac{2}{3} \left[-\frac{1}{t} \right]_4^7 = \frac{2}{3} \left(-\frac{1}{7} + \frac{1}{4} \right) = \frac{1}{14}$

Ans.**Do yourself - 7 :****Evaluate :**

1. $\lim_{n \rightarrow \infty} \left[\frac{1}{n+2.1} + \frac{1}{n+2.2} + \frac{1}{n+2.3} + \dots + \frac{1}{3n} \right]$

2. $\lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{1}{\sqrt{n^2 - r^2}}$

3. The value of $\lim_{n \rightarrow \infty} \sum_{r=1}^{r=4n} \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r}+4\sqrt{n})^2}$ is equal to

(A) $\frac{1}{35}$

(B) $\frac{1}{14}$

(C) $\frac{1}{10}$

(D) $\frac{1}{5}$

4. The value of $\lim_{n \rightarrow \infty} \sum_{r=0}^{2n-1} \frac{n^2}{n^2 + 4r^2}$ is :
- (A) $\frac{1}{2} \tan^{-1}(2)$ (B) $\frac{1}{2} \tan^{-1}(4)$ (C) $\tan^{-1}(4)$ (D) $\frac{1}{4} \tan^{-1}(4)$
5. $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \dots + \frac{n}{(2n-1)^2} \right]$ is equal to :
- (A) $\frac{1}{2}$ (B) 1 (C) $\frac{1}{3}$ (D) $\frac{1}{4}$
6. If $L = \lim_{n \rightarrow \infty} \left(\frac{n+2}{n^2+n+1} + \frac{n+4}{n^2+2n+4} + \frac{n+6}{n^2+3n+9} + \dots + \frac{5}{7n} \right)$, then e^L is equal to -
- (A) 1 (B) $\frac{1}{7}$ (C) 7 (D) e
7. $\lim_{n \rightarrow \infty} \left\{ \sin\left(\frac{\pi}{2n}\right) \cdot \sin\left(\frac{2\pi}{2n}\right) \sin\left(\frac{3\pi}{2n}\right) \dots \sin\left(\frac{(n-1)\pi}{2n}\right) \right\}^{1/n}$ is equal to -
- (A) $\sin 1$ (B) $\frac{1}{2}$ (C) $\ln 2$ (D) $e^{1/2}$
8. If $\lambda = \lim_{n \rightarrow \infty} \frac{\left(\sum_{r=1}^n r \right) \left(\sum_{r=1}^n r^2 \right) \left(\sum_{r=1}^n r^3 \right)}{\sum_{r=1}^n r^8}$, then the value of 8λ is

6. ESTIMATION OF DEFINITE INTEGRAL :

- (a) If $f(x)$ is continuous in $[a, b]$ and its range in this interval is $[m, M]$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

Illustration 24: Prove that $4 \leq \int_1^3 \sqrt{3+x^3} dx \leq 2\sqrt{30}$

Solution : Since the function $f(x) = \sqrt{3+x^3}$ increases monotonically on the interval $[1, 3]$, $m = 2$, $M = \sqrt{30}$, $b - a = 2$.

$$\text{Hence, } 2.2 \leq \int_1^3 \sqrt{3+x^3} dx \leq 2\sqrt{30} \Rightarrow 4 \leq \int_1^3 \sqrt{3+x^3} dx \leq 2\sqrt{30}$$

Ans.

(b) If $f(x) \leq \phi(x)$ for $a \leq x \leq b$ then $\int_a^b f(x) dx \leq \int_a^b \phi(x) dx$

Illustration 25: Prove that $\frac{\pi}{6} \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \frac{\pi}{4\sqrt{2}}$

Solution : Since $4 - x^2 \geq 4 - x^2 - x^3 \geq 4 - 2x^2 > 0 \quad \forall x \in [0, 1]$

$$\sqrt{4-x^2} \geq \sqrt{4-x^2-x^3} \geq \sqrt{4-2x^2} > 0 \quad \forall x \in [0, 1]$$

$$\Rightarrow 0 < \frac{1}{\sqrt{4-x^2}} \leq \frac{1}{\sqrt{4-x^2-x^3}} \leq \frac{1}{\sqrt{4-2x^2}} \quad \forall x \in [0, 1]$$

$$\Rightarrow \int_0^1 \frac{dx}{\sqrt{4-x^2}} \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \int_0^1 \frac{dx}{\sqrt{4-2x^2}} \quad \forall x \in [0, 1]$$

$$\Rightarrow \left[\sin^{-1} \frac{x}{2} \right]_0^1 \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \frac{1}{\sqrt{2}} \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1 \Rightarrow \frac{\pi}{6} \leq \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} \leq \frac{\pi}{4\sqrt{2}} \text{ Ans.}$$

(c) $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$

Illustration 26: Prove that $\left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| < 10^{-7}$

Solution : To find $I = \left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| \leq \int_{10}^{19} \left| \frac{\sin x}{1+x^8} \right| dx$ (i)

Since $|\sin x| \leq 1$ for $x \geq 10$

The inequality $\left| \frac{\sin x}{1+x^8} \right| \leq \frac{1}{1+x^8}$ (ii)

also, $10 \leq x \leq 19$

$$\Rightarrow 1+x^8 > 10^8$$

$$\Rightarrow \frac{1}{1+x^8} < \frac{1}{10^8} \text{ or } \frac{1}{|1+x^8|} < 10^{-8} \text{ (iii)}$$

from (ii) and (iii) ;

$$\left| \frac{\sin x}{1+x^8} \right| < 10^{-8}$$

$$\left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| < \int_{10}^{19} 10^{-8} dx$$

$$\therefore \left| \int_{10}^{19} \frac{\sin x}{1+x^8} dx \right| < (19-10) \cdot 10^{-8} < 10^{-7}$$

Ans.

Illustration 27 : If $f(x)$ is integrable function such that $|f(x) - f(y)| \leq |x^2 - y^2|$, $\forall x, y \in [a, b]$ then prove

$$\text{that } \left| \int_a^b \frac{f(x) - f(a)}{x + a} dx \right| \leq \frac{(a - b)^2}{2}.$$

Solution :

$$\begin{aligned} \text{Given, } \left| \int_a^b \frac{f(x) - f(a)}{x + a} dx \right| &\leq \int_a^b \left| \frac{f(x) - f(a)}{x + a} \right| dx \\ &\leq \int_a^b \left| \frac{x^2 - a^2}{x + a} \right| dx = \int_a^b |x - a| dx = \int_a^b (x - a) dx = \frac{(a - b)^2}{2} \end{aligned}$$

(d) If $f(x) \geq 0$ on the interval $[a, b]$, then $\int_a^b f(x) dx \geq 0$.

Illustration 28 : If $f(x)$ is a continuous function such that $f(x) \geq 0 \forall x \in [2, 10]$ and $\int_4^8 f(x) dx = 0$, then find $f(6)$.

Solution : $f(x)$ is above the x-axis or on the x-axis for all $x \in [2, 10]$. If $f(x)$ is greater than zero for any sub interval of $[4, 8]$, then $\int_4^8 f(x) dx$ must be greater than zero.

$$\begin{aligned} \text{But } \int_4^8 f(x) dx = 0 &\Rightarrow f(x) = 0 \forall x \in [4, 8] \\ &\Rightarrow f(6) = 0. \end{aligned}$$

Do yourself - 8 :

1. Prove that $4 \leq \int_1^3 \sqrt{3 + x^2} dx \leq 4\sqrt{3}$
2. Prove that $\frac{\pi}{4} \leq \int_0^{2\pi} \frac{dx}{5 + 3 \sin x} \leq \pi$.
3. Show that $\frac{3}{5} (2^{1/3} - 1) \leq \int_0^1 \frac{x^4}{(1 + x^6)^{2/3}} dx \leq 1$
4. Prove the inequalities :

(a) $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4 - x^2 - x^3}} < \frac{\pi\sqrt{2}}{8}$

(b) $2e^{-1/4} < \int_0^2 e^{x^2 - x} dx < 2e^2$
5. Prove the inequalities :

(a) $\frac{1}{3} < \int_0^1 x^{(\sin x + \cos x)^2} dx < \frac{1}{2}$

(b) $\frac{1}{2} \leq \int_0^2 \frac{dx}{2 + x^2} \leq \frac{5}{6}$

Miscellaneous Illustrations :

Illustration 29: Evaluate : $\int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx$

Solution : Let $I = \int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx$ (i)

$$= \int_0^\pi \frac{(\pi-x)^3 \cos^4(\pi-x) \sin^2(\pi-x) dx}{\pi^2 - 3\pi(\pi-x) + 3(\pi-x)^2} \quad (\text{By. Prop.})$$

$$= \int_0^{\pi} \frac{(\pi^3 - x^3 - 3\pi^2 x + 3\pi x^2) \cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx \quad \dots\dots\dots (ii)$$

Adding (i) and (ii) we have

$$2I = \int_0^{\pi} \frac{(\pi^3 - 3\pi^2x + 3\pi x^2) \cos^4 x \sin^2 x}{(\pi^2 - 3\pi x + 3x^2)} dx$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \cos^4 x \sin^2 x dx \Rightarrow 2I = 2\pi \int_0^{\pi/2} \cos^4 x \sin^2 x dx$$

$$\therefore I = \pi \int_0^{\pi/2} \cos^4 x \sin^2 x \, dx$$

Using walli's formula, we get $I = \pi \frac{(3.1)(1)}{6.4.2} \frac{\pi}{2} = \frac{\pi^2}{32}$

Ans.

Illustration 30: Let f be a polynomial function such that $f(x)f(y) + 2 = f(x) + f(y) + f(xy) \quad \forall x, y \in \mathbb{R}$ with $f(0) = 1$ and $f(1) = 2$ find $f(x)$ and show that $3 \int f(x)dx - x(f(x) + 2)$ is a constant.

Solution : We have $f(x)f(y) + 2 = f(x) + f(y) + f(xy)$

Putting $x = 1$ & $y = 1$

then $f(1)f(1) + 2 = 3f(1)$

we get $f(1) = 1, 2$

$$f(1) \neq 1$$

then $f(1) = 2$

Replacing y by $\frac{1}{x}$ in (1) then

$$f(x)f\left(\frac{1}{x}\right)+2=f(x)+f\left(\frac{1}{x}\right)+f(1) \Rightarrow f(x)f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$$

Illustration 33: If $f(x) = \begin{cases} 1-|x|, & |x| \leq 1 \\ |x|-1, & |x| > 1 \end{cases}$, and $g(x) = f(x-1) + f(x+1)$. Find the value of $\int_{-3}^5 g(x) dx$.

Solution : Given,

$$f(x) = \begin{cases} -x-1, & x < -1 \\ 1+x, & -1 \leq x < 0 \\ 1-x, & 0 \leq x \leq 1 \\ x-1, & x > 1 \end{cases}; \quad f(x-1) = \begin{cases} -x, & x-1 < -1 \Rightarrow x < 0 \\ x, & -1 \leq x-1 < 0 \Rightarrow 0 \leq x < 1 \\ 2-x, & 0 \leq x-1 \leq 1 \Rightarrow 1 \leq x \leq 2 \\ x-2, & x-1 > 1 \Rightarrow x > 2 \end{cases}$$

Similarly

$$f(x+1) = \begin{cases} -x-2, & x+1 < -1 \Rightarrow x < -2 \\ x+2, & -1 \leq x+1 < 0 \Rightarrow -2 \leq x < -1 \\ -x, & 0 \leq x+1 \leq 1 \Rightarrow -1 \leq x \leq 0 \\ x, & x+1 > 1 \Rightarrow x > 0 \end{cases}$$

$$\Rightarrow g(x) = f(x-1) + f(x+1) = \begin{cases} -2x-2, & x < -2 \\ 2, & -2 \leq x < -1 \\ -2x, & -1 \leq x \leq 0 \\ 2x, & 0 < x < 1 \\ 2, & 1 < x \leq 2 \\ 2x-2, & 2 < x \end{cases}$$

Clearly $g(x)$ is even,

$$\text{Now } \int_{-3}^5 g(x) dx = 2 \int_0^3 g(x) dx + \int_3^5 g(x) dx$$

$$= 2 \left(\int_0^1 2x dx + \int_1^2 2 dx + \int_2^3 (2x-2) dx \right) + \int_3^5 (2x-2) dx = 24$$